

Brain4FMs: A Benchmark of Foundation Models on Electrical Brain Signals

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Abstract

Brain foundation models (BFMs) are advancing neurotechnology by learning transferable representations from neural signals, with broad potential in clinical diagnosis and neuroscience research. Their development relies on large-scale pretraining corpora of electrical brain signals, including scalp electroencephalography (EEG) and intracranial EEG (iEEG). However, existing BFM benchmarks primarily focus on EEG, cover only a limited subset of models, and provide limited analysis beyond downstream performance. We introduce Brain4FMs, the first unified benchmark, to our knowledge, for jointly evaluating BFMs on EEG and iEEG. It integrates 17 representative models and 21 public datasets across clinical diagnosis, sleep staging, communication, and affective computing. Brain4FMs is open and plug-and-play, with dataset-aware preprocessing, cross-subject evaluation, heterogeneous multichannel handling, and standardized downstream adaptation workflows. The benchmark reveals performance variation across tasks, signal modalities, and adaptation protocols, with no single BFM consistently dominating all evaluation scenarios. To better understand these behaviors, we further conduct exploratory analyses of model-specific properties of spatial, spectral, and discrete representations.

Code — <https://github.com/wajtsq/Brain4FMs>

Introduction

Neuroscience is fundamental to understanding the brain and translating insights into societal impact. Electrical brain signals, including scalp electroencephalography (EEG) and intracranial EEG (iEEG), provide millisecond-level measurements of neural activity. EEG supports non-invasive, scalable recording, whereas iEEG offers higher-fidelity and more spatially precise measurements of cortical or deep-brain activity but is restricted to clinically indicated patients due to its invasive nature (Parvizi and Kastner 2018). Together, these complementary modalities support a wide range of applications, including neurological disease detection, sleep staging, neural communication, and affective computing.

Deep learning has improved neural signal decoding (Li et al. 2021; Jing et al. 2023), but its generalization is often

limited by costly annotations, inter-subject variability, and distribution shifts across devices or tasks. To overcome these limitations, self-supervised learning (SSL) leverages unlabeled neural data for representation learning (Henaff 2020). This shift parallels advances in Natural Language Processing and Computer Vision, where large-scale pretraining enabled foundation models. A similar transition is emerging in neuroscience, with Brain Foundation Models (BFMs) serving as universal encoders that learn robust representations and support adaptation to downstream tasks.

Despite rapid progress, existing BFM benchmarks remain largely restricted to EEG and evaluate only a limited subset of available BFMs, leaving their performance on iEEG recordings unclear (Xiong et al. 2025; Wu et al. 2025; Lu et al. 2026). Duan et al. (2026) advance standardized iEEG evaluation through epilepsy, but focus on task-specific baselines rather than existing BFMs. Figure 6(a) summarizes the complementary benchmark scopes. Thus, these limitations motivate a unified cross-subject benchmark for comparing BFMs across EEG and iEEG. To bridge these gaps, we introduce Brain4FMs, a standardized benchmark for evaluating BFMs on EEG/iEEG classification. The evaluation centers around three questions: 1) How do current BFMs perform across tasks and modalities? 2) How does adaptation change their transferability? 3) What structural patterns can be observed in their representations? Our main contributions are:

- 1) **A Unified EEG/iEEG Benchmark.** We introduce Brain4FMs, an open benchmark that integrates 17 representative BFMs and 21 public EEG/iEEG datasets covering 26 classification settings in clinical diagnosis, sleep staging, communication, and affective computing.
- 2) **Standardized Evaluation Protocol.** We establish an extensible cross-subject evaluation framework that standardizes downstream procedures while accommodating model- and modality-specific requirements.
- 3) **Extensive Empirical Evaluation.** We report substantial variation in BFM performance and modality-associated behavior under standardized evaluation, complemented by exploratory analyses of model representations.

Benchmark Design

To support standardized and reproducible evaluation, we introduce Brain4FMs, a comprehensive benchmark for

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EEG/iEEG classification with BFMs. It unifies data processing, model integration, and cross-subject evaluation, with plug-and-play interfaces for adding new models and datasets.

Dataset coverage

We select well-established datasets that support cross-subject evaluation and are widely used in brain-signal research. Dataset selection balances task diversity, recording modality, and practical accessibility, while avoiding overlap with the reported pretraining corpora of evaluated BFMs whenever possible. Brain4FMs covers 21 public datasets and 26 classification settings, with five datasets contributing two settings each. Table 1 summarizes the benchmark datasets, with complete statistics provided in the supplement.

Disease Diagnosis. Neurological disorders are among the leading causes of disability and mortality worldwide (Li et al. 2025b). To support clinically relevant evaluation, Brain4FMs covers epilepsy, drug-resistant epilepsy (DRE), depression, Major Depressive Disorder (MDD), Schizophrenia (SZ), Alzheimer’s Disease (AD), Parkinson’s disease (PD) and Attention Deficit Hyperactivity Disorder (ADHD). These tasks use clinically grounded labels for realistic evaluation.

Sleep Staging. Sleep staging supports sleep disorder diagnosis, but manual overnight scoring is labor-intensive, motivating automated models that capture long-range temporal dependencies and rich time-frequency patterns.

Communication. Communication tasks map neural activity to external outputs for intent expression. Typical tasks include Motor Imagery (MI), Motor Execution (ME), and multimodal neural decoding, which aims to infer visual, auditory, or semantic information from brain signals.

Affective Computing. Affective computing infers internal emotional and cognitive states from EEG/iEEG. Core tasks include emotion recognition and mental workload estimation (MW), which involve temporal, spectral, and spatial patterns of affective and cognitive demand across diverse settings.

Model Interfaces

As summarized in Table 2, Brain4FMs includes 17 representative BFMs spanning diverse architectures and pretraining objectives. We include BFMs from peer-reviewed studies or widely cited preprints with publicly available code and pretrained checkpoints. All models use the same dataset-specific preprocessing pipeline and standardized downstream evaluation procedures while retaining each model’s original architecture, thereby assessing system-level transfer under controlled adaptation. Given an EEG or iEEG sample $X \in \mathbb{R}^{C \times N}$, where C and N denote the numbers of channels and temporal samples, respectively, each encoder produces a latent representation for downstream prediction. Plug-and-play interfaces further support new models. We group the included BFMs by their primary pretraining mechanisms.

Contrastive-based Methods. Contrastive methods learn representations by aligning related signal views or temporal segments while separating mismatched examples. SppEEGNet (Li and Metsis 2022) and BIOT (Yang, Westover, and

Sun 2023) construct positive pairs through signal augmentations, whereas Bendr (Kostas, Aroca-Ouellette, and Rudzicz 2021) and MBrain (Cai et al. 2023) adopt predictive or spatiotemporal contrastive objectives.

Name	Signal	Task	Cat.
CHBMIT (Guttag 2010)	EEG	Epilepsy	2
MAYO (Nejedly et al. 2020)	iEEG	DRE	2
FNUSA (Nejedly et al. 2020)	iEEG	DRE	2
HUP (Bernabei et al. 2023)	iEEG	DRE	2/2
SWEC (Carzaniga et al. 2025)	iEEG	DRE	2
Dep (Cavanagh 2021)	EEG	Depression	2/3
MDD-64 (Mumtaz 2016)	EEG	MDD	2
SZ-28 (Olejarczyk and Jernajczyk 2017)	EEG	SZ	2
UCSD (Rockhill et al. 2021)	EEG	PD	2/2
ADFD (Miltiadous et al. 2023)	EEG	AD	2
ADHD-Adult (Bajestani et al. 2023)	EEG	ADHD	2
ADHD-Child (Nasrabadi et al. 2020)	EEG	ADHD	2
ISRUC-G1 (Khalighi et al. 2016)	EEG	Sleep Stage	5
SleepEDF (Kemp et al. 2000)	EEG	Sleep Stage	5
EEGMMIDB (Goldberger et al. 2000)	EEG	ME	4/4
BCI-2a (Brunner et al. 2008)	EEG	MI	4
Chisco (Zhang et al. 2024b)	EEG	Concept	39/39
Cogitate-CF (Seedat et al. 2025)	iEEG	Visual	2
DEAP (Koelstra et al. 2011)	EEG	Emotion	4
SEED-IV (Zheng et al. 2018)	EEG	Emotion	4
EEGMat (Zyma et al. 2019)	EEG	MW	2

Table 1: Benchmark datasets by signal type, task, and number of classes (Cat.); "/" separates two evaluation settings.

Name	Strategy	Param	Modality	Domain
Contrastive-based Method				
SppEEGNet	Aug.	138 K	EEG	time
BIOT	Aug.	3.19 M	EEG	time, frequency
Bendr	CPC	3.97 M	EEG	time, frequency
MBrain	CPC	8.34 M	EEG/iEEG	time, frequency, space
Generative-based Method				
Brant	AE	196.10 M	iEEG	time, frequency, space
BFM	AE	708.96 M	EEG	time
BrainBERT	AE	43.18 M	iEEG	time, frequency
CBraMod	AE	4.88 M	EEG	time, frequency, space
NeuroGPT	AR	79.62 M	EEG	time, frequency
LaBraM	AE	5.80 M	EEG	time, frequency, space
BrainWave	AE	102.13 M	EEG+iEEG	time, frequency, space
REVE	AE	69.19 M	EEG	time, space
BrainOmni	AE	32.71 M	EEG	time, frequency, space
CodeBrain	AE	15.17 M	EEG	time, frequency, space
Other Advanced Methods				
EEGPT	G & C	51.04 M	EEG	time, space
NeuroLM	G & Adv.	169.60 M	EEG	time, frequency
MVPFormer	G & C	73.59 M	iEEG	time, frequency, space

Table 2: Overview of models by SSL strategy, encoder size, pretraining modality, and feature domain. C/G/O/Adv. indicate contrastive/generative/other/adversarial methods.

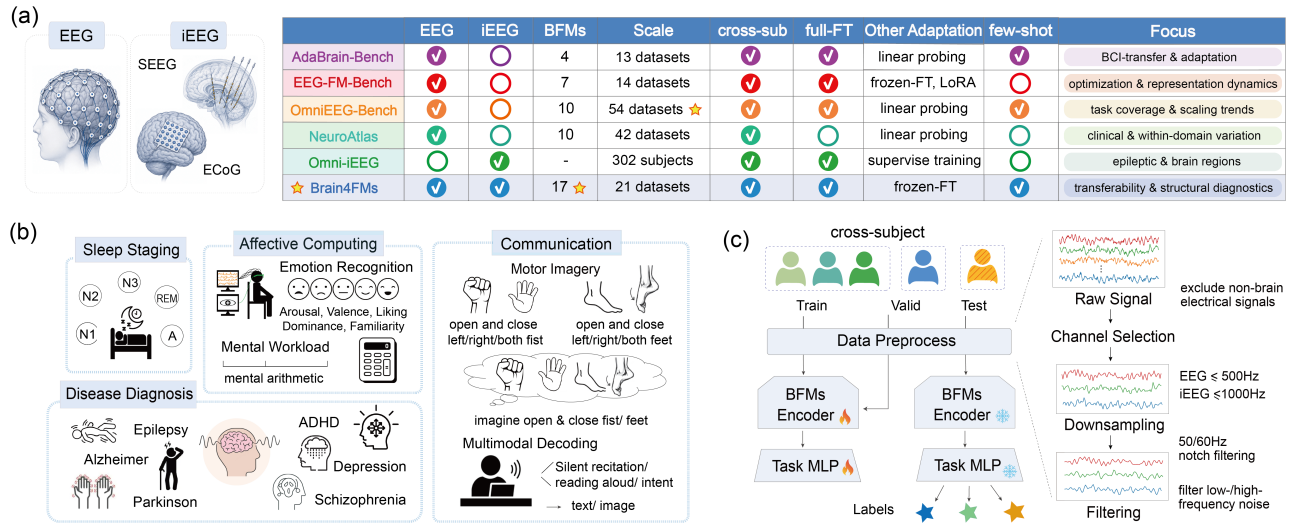


Figure 1: Overview of Brain4FMs. (a) Scope of related work by signal modality, benchmark scale, evaluation protocol, adaptation setting, and analysis focus. (b) EEG/iEEG acquisition across diverse clinical and human-centered scenarios. (c) Dataset-aware preprocessing and subject-disjoint cross-validation under a standardized evaluation framework.

Generative-based Methods. These methods learn representations by reconstructing masked signals, predicting latent targets, or modeling neural signals as token sequences. BrainBERT, Brant, BrainWave, CBraMod, REVE, and CodeBrain primarily employ masked or structured reconstruction (Wang et al. 2023; Zhang et al. 2023a; Yuan et al. 2024; Wang et al. 2024b; Ouahidi et al. 2025; Ma et al. 2025). NeuroGPT (Cui et al. 2024) adopts autoregressive prediction, whereas LaBraM, BFM, and BrainOmni incorporate discrete tokenization or codebook-based modeling (Jiang, Zhao, and Lu 2024; Bayazi et al. 2024; Xiao et al. 2025).

Other Advanced Methods. Beyond contrastive and generative SSL, BFMs also explore predictive, hybrid, adversarial, and instruction-tuning strategies. EEGPT and MVPFormer combine generative and contrastive objectives, whereas NeuroLM combines generative and adversarial learning (Wang et al. 2024a; Carzaniga et al. 2025; Jiang et al. 2024).

Evaluation Protocol

Brain4FMs focuses on classification tasks, which aim to divide the input EEG/iEEG samples into pre-defined categories. The pipeline consists of two components: dataset preprocessing and downstream evaluation (Figure 6(c)).

Dataset preprocessing. We standardize data organization and preprocessing principles rather than enforcing identical signal-processing parameters across datasets. The pipeline applies dataset-appropriate bandpass and notch filtering, downsampling, event-aligned window segmentation, channel selection, and per-channel z-score normalization. We use group-wise cross-validation with approximately 3:1:1 cross-subject training, validation, and test splits, ensuring that recordings from the same subject never appear across splits. For iEEG with patient-specific channel counts, compatible models use patient-wise batching to preserve native

channel structures without cross-subject alignment.

Downstream Evaluation. Each pretrained BFM serves as the backbone with its model-compatible classification interface. Full-parameter fine-tuning is used as the primary setting, while frozen-encoder and prototype-based low-label evaluations are included as complementary analyses. For each model-dataset pair, we search the encoder learning rate (LR) over $\{10^{-4}, 5 \times 10^{-5}, 10^{-5}, 5 \times 10^{-6}, 10^{-6}\}$, with the classification-head LR set to ten times the encoder rate. Within each fold, checkpoints are selected by validation loss, and the learning rate is selected using the primary validation metric: AUROC for binary tasks and Cohen’s κ for multiclass tasks. Final performance is averaged across test folds, with selected LR, complete metrics, and training details such as the optimizer, batch size, and schedule reported in the supplement.

Empirical Results

Q1: How do current BFMs perform across tasks and modalities? To answer this question, we report the main results under full-FT, followed by analyses of task performance, EEG and iEEG modalities, and pretraining paradigms.

Task Performance

We compare BFMs with supervised baselines selected for each task under fully matched experimental settings. The reference models include SPaRCNet (Jing et al. 2023) for epilepsy and ADHD, DeprNet for depression (Seal et al. 2021), LocCNN (Choi et al. 2026) for AD and PD, CC-NSE for sleep staging (Li et al. 2021), and MSCARNet for communication and affective computing tasks (Avola et al. 2024). As shown in Tables 3 and 4, BFMs exhibit stronger cross-subject performance on epilepsy and sleep staging. Supervised baselines remain competitive but typically outperform only a subset of BFMs. On depression datasets, DeprNet

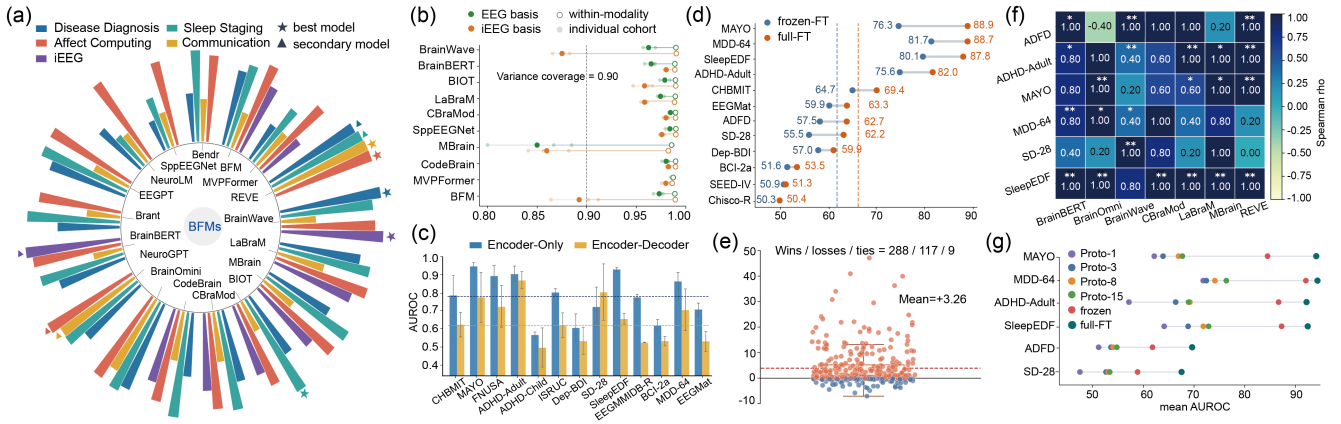


Figure 2: Empirical results and downstream adaptation analyses. (a) Overall BFM performance, averaged by task using AUROC. (b) Bidirectional EEG/iEEG subspace coverage. (c) AUROC comparison of encoder-only and encoder-decoder NeuroGPT variants. (d-e) AUROC gains of full-FT over frozen-FT across BFMs. (f) Pair-level trends across Proto-1/3/8/15, * $q < 0.05$, ** $q < 0.01$ by the BH-corrected Page test. (g) Mean AUROC under prototype transfer, frozen-FT, and full-FT.

achieves moderate AUROC but low accuracy, indicating limited robustness. On PD, LocCNN outperforms most BFMs, although cross-subject performance remains unstable. Overall, BFMs surpass the selected supervised references on most clinical tasks, showing promise for cross-subject transfer.

Performance within affective computing and communication is highly task dependent. Cross-subject concept decoding, visual classification and emotion recognition remain particularly challenging because of pronounced non-stationarity and inter-subject variability (Lotte et al. 2018). On these tasks, the BCI-tailored supervised baseline MSCARNet outperforms most BFMs, although models such as NeuroGPT remain competitive on selected datasets. In contrast, MI, ME, and MW yield stronger results for several BFMs: REVE and NeuroGPT achieve competitive accuracy and AUROC, while BrainOmni performs strongly on EEGMMIDB. These observations suggest that current BFMs transfer more consistently to motor-related classification than to higher-level semantic or affective decoding under the cross-subject setting. NeuroLM, despite pretraining on emotion-related data, does not consistently generalize across subjects.

EEG and iEEG modalities

The benchmark reveals a modality-matched transfer trend. On the epilepsy datasets, iEEG-pretrained models generally perform better on intracranial datasets, whereas EEG-pretrained models are more competitive on scalp EEG. For example, BrainBERT performs strongly on FNUSA and HUP-ECOG but its relative advantage is smaller on CHBMIT. LaBraM performs well on EEG and single-channel iEEG datasets but falls behind supervised baselines on the patient-specific HUP datasets. BrainWave, jointly pretrained on EEG and iEEG, remains competitive across both modalities, consistent with its controlled finding that joint EEG-iEEG pretraining outperforms EEG-only and iEEG-only variants.

Motivated by this trend, we analyzed cross-modal subspace overlap and mixed-space expansion in ten pretrained

BFMs using matched 2-s epochs from six EEG and six multi-channel iEEG cohorts. From each cohort, we sampled 2,000 label-balanced epochs and centered representations within the cohort before pooling, which yielded 12,000 samples per modality. To assess robustness to EEG cohort composition, we repeated the analysis with three random EEG-cohort selections. For each modality, we extracted the PCA subspace explaining 99% of variance. Directional coverage $C_{S \rightarrow T}$ is the fraction of target variance captured by the source subspace. We estimated coverage on complementary sample halves and computed within-modality coverage in the same way. Modality-specific residual structure is defined as $G = C_{\text{within}} - C_{\text{cross}}$, with larger values indicating more modality-specific variance.

	HUP-ECOG		FNUSA		CHBMIT	
Model	AUROC	Acc	AUROC	Acc	AUROC	Acc
MBrain	.57±.09	.61±.12	.91±.08	.87±.08	.75±.08	.71±.12
BIOT	.54±.08	.61±.12	.87±.07	.83±.08	.56±.08	.29±.05
SppEEGNet	.44±.07	.52±.09	.82±.09	.84±.09	.43±.05	.60±.06
BrainWave	.88±.07	.80±.11	.92±.05	.89±.06	.90±.03	.80±.12
BrainBERT	.81±.12	.73±.09	.90±.04	.89±.04	.78±.06	.75±.10
BFM	.72±.11	.71±.10	.78±.12	.69±.11	.74±.05	.72±.03
CodeBrain	.67±.05	.71±.06	.83±.10	.85±.06	.60±.08	.74±.10
CBraMod	.64±.09	.58±.10	.71±.13	.78±.08	.69±.04	.70±.08
LaBraM	.51±.06	.64±.13	.89±.08	.82±.10	.75±.10	.73±.10
SPaRCNet	.56±.01	.59±.09	.85±.10	.82±.11	.61±.10	.66±.05

Table 3: Primary epilepsy metrics for representative models.

The effective dimensionality of representations is primarily model dependent, with broadly consistent distributions across datasets within each model. Across all ten BFMs, the dominant EEG and iEEG subspaces are largely shared: mean bidirectional coverage exceeds 0.85 for every model, indicating that each modality’s principal variance is largely captured

	ADHD-Adult		ADHD-Child		ADFD		MDD-64		SZ-28		SleepEDF		UCSD-ON	
Model	AUROC	Acc	AUROC	Acc	AUROC	Acc	AUROC	Acc	AUROC	Acc	AUROC	Acc	AUROC	Acc
MBrain	.95±.03	.92±.04	.69±.11	.63±.08	.53±.12	.50±.10	.93±.08	.87±.11	.58±.10	.58±.10	.95±.01	.78±.04	.49±.15	.46±.14
Bendr	.62±.03	.61±.04	.53±.02	.51±.01	.52±.02	.52±.01	.85±.07	.78±.05	.49±.05	.50±.03	.91±.01	.69±.06	.51±.05	.49±.05
BrainOmni	.96±.02	.91±.02	.71±.09	.62±.07	.80±.07	.71±.04	.94±.07	.85±.09	.69±.15	.69±.12	.87±.00	.63±.03	.53±.14	.49±.12
BrainWave	.96±.03	.91±.04	.78±.04	.71±.04	.74±.05	.68±.05	.94±.03	.85±.02	.88±.07	.81±.06	.89±.02	.67±.05	.69±.10	.53±.08
REVE	.96±.02	.91±.02	.79±.07	.71±.04	.84±.07	.77±.08	.97±.04	.88±.09	.81±.13	.75±.13	.94±.02	.75±.04	.63±.19	.56±.13
LaBraM	.95±.04	.91±.04	.66±.12	.64±.08	.83±.10	.75±.07	.96±.04	.87±.08	.54±.07	.54±.04	.93±.01	.76±.04	.51±.22	.45±.11
CBraMod	.95±.02	.92±.04	.64±.05	.64±.06	.55±.06	.59±.04	.90±.13	.84±.12	.54±.14	.58±.05	.93±.01	.73±.04	.60±.12	.49±.08
BrainBERT	.74±.05	.69±.05	.55±.03	.56±.03	.59±.05	.58±.05	.92±.07	.85±.07	.73±.11	.66±.15	.91±.02	.68±.07	.51±.05	.51±.06
EEGPT	.96±.03	.91±.04	.67±.13	.64±.10	.81±.05	.72±.05	.94±.05	.87±.08	.56±.21	.56±.11	.91±.00	.68±.03	.51±.21	.48±.17
$SL^\dagger$	.94±.02	.89±.03	.66±.05	.62±.03	.76±.05	.70±.05	.91±.06	.78±.04	.78±.15	.74±.09	.84±.02	.33±.04	.55±.23	.52±.18

Table 4: Primary performance metrics on clinical tasks. We show a subset of BFM and $SL^\dagger$ denotes supervised baselines.

by the other modality’s subspace. This pattern is robust to EEG cohort selection, with highly consistent model rankings across the three draws. However, the overlap can be asymmetric. For BrainWave, $C_{EEG \rightarrow iEEG} = 0.96$, compared with 0.88 in the reverse direction, indicating modality-specific residuals. Moreover, the residual gap G is positively correlated with macro-averaged AUROC across the 12 cohorts in each draw ($\rho = 0.86$ -0.90, all $p < 0.01$). This association is stronger for iEEG tasks ($\rho = 0.81$ -0.87) than for EEG tasks ($\rho = 0.66$ -0.75). Together, these results suggest that BFMs learn a largely shared EEG/iEEG representational backbone while retaining modality-specific residual structure associated with higher frozen-transfer performance.

Pretraining Paradigms

Beyond data modality, we also observe that SSL objectives are associated with different transfer behaviors. Among contrastive BFMs, MBrain shows the strongest overall performance on clinically relevant tasks. Compared with other models, MBrain uniquely combines multichannel CPC with graph-based modeling. Within generative BFMs, most models adopt autoencoding (AE) rather than autoregressive (AR) pretraining. AR models predict tokens sequentially under a causal context (Liu et al. 2021), whereas AE can aggregate information across the full input window. The latter may be better aligned with downstream classification, which typically relies on window-level discrimination rather than sequence generation. The comparison between the encoder-only NeuroGPT-E and the encoder-decoder NeuroGPT-D further supports this observation. As shown in Figure 2(c), NeuroGPT-E consistently outperforms NeuroGPT-D across epilepsy, sleep staging, MW, and MI tasks, suggesting that downstream classification relies primarily on encoder representations, with limited additional benefit from the decoder.

Downstream Adaptation

Q2: How does adaptation change their transferability?

To address this question, we consider two complementary settings: frozen-encoder transfer, which isolates the benefit of pretrained representations, and prototype-based transfer, which evaluates representations under limited supervision.

Frozen-Encoder Transfer

We compare frozen-encoder fine-tuning (frozen-FT) with full fine-tuning (full-FT). Frozen-FT fixes the pretrained encoder and optimizes only the classifier, directly evaluating the transferability of frozen representations. Full-FT instead updates both the encoder and classifier, allowing us to quantify the additional gain from adapting the pretrained encoder to the target task. To ensure a controlled comparison, frozen-FT uses the same classifier-head LR selected for the corresponding full-FT setting, with detailed results reported in the supplementary material. After averaging AUROC across cross-validation folds for each model–dataset pair, we obtain paired observations ($n=414$). Full-FT improves over frozen-FT by 3.26 AUROC points on average, with significant gains under both the Wilcoxon signed-rank test ($p = 3.49 \times 10^{-25}$) and the sign test ($p = 5.96 \times 10^{-17}$). These results show that transfer performance depends not only on pretrained representations but also on their downstream adaptation.

The gain from full-FT is not uniform: most model–dataset pairs show moderate improvements, while a few benefit substantially from encoder updates. Representative examples include Bendr on MAYO (+47.2 AUROC) and BIOT on MDD-64 (+36.4), as shown in Figure 2(d). In these cases, frozen representations provide a useful initialization, but full-FT is needed to reshape the feature space and form more effective task-specific decision boundaries. Full-FT is not always better than frozen-FT. In a few cases, frozen-FT achieves slightly higher AUROC. However, these cases generally correspond to weak or unstable transfer performance rather than strong frozen representations. This suggests that the benefit of full-FT may be limited or unstable on difficult tasks. Despite these differences, frozen-FT and full-FT show strong ranking consistency across model-dataset pairs (Spearman $\rho = 0.82$, $p = 2.29 \times 10^{-91}$). Thus, frozen-FT remains an important indicator of BFM transferability: it largely preserves the relative ranking among models, while full-FT further raises the achievable performance ceiling through encoder adaptation.

Prototype-Based Few-Shot Transfer

Although most BFMs are not explicitly trained for few-shot inference, limited labeled brain-signal data make this set-

ting practically important. We therefore adopt a parameter-free prototype-based protocol that classifies each query by comparing its representation with class prototypes. For each class, we sample a small support set and define its prototype as the mean frozen representation: $\mathbf{p}_c = \frac{1}{|\mathcal{S}_c|} \sum_{i \in \mathcal{S}_c} f_\theta(x_i)$, where $\mathcal{S}_c$ denotes the sampled support set for class c , and $f_\theta(\cdot)$ is the frozen BFM encoder. A test sample x is assigned to the class whose prototype has the highest cosine similarity averaged across channels: $\hat{y} = \arg \max_c \text{sim}(f_\theta(x), \mathbf{p}_c)$. This protocol evaluates whether the frozen representation supports class discrimination under limited supervision without updating model parameters.

Figure 2(g) reports results for seven high-performing BFMs on six datasets under 1-, 3-, 8-, and 15-shot settings, with each fold repeated five times. Prototype-based transfer remains below full-FT in absolute performance, but generally improves as the support set grows. Proto-15 outperforms Proto-1 in 95% of the 42 model-dataset pairs, with an average gain of 6.68 AUROC points. Across the ordered 1/3/8/15-shot settings, most model-dataset pairs show positive trend correlations, with a mean Spearman correlation of $\rho = 0.73$. We assess these trends using the Page test and control the false discovery rate with the Benjamini-Hochberg procedure, as shown in Figure 2(f). The pair-level results show that the gains are heterogeneous: only a subset of model-dataset pairs retain statistically significant positive trends after correction.

Prototype performance also moderately agrees with the model ordering under full-FT. Across Proto-1, Proto-3, Proto-8, and Proto-15, the Spearman correlations with full-FT range from 0.64 to 0.68. The agreement is stronger on MAYO and SleepEDF, where the correlations range from 0.75-0.79 and 0.79-0.86, respectively. Frozen-FT generally remains stronger than prototype-based transfer, including at 15 shots. However, prototype-based transfer outperforms frozen-FT for seven of the 42 model-dataset pairs in at least two shot settings, including LaBraM, BrainWave, and CBraMod on SZ-28; MBrain, CBraMod, and BrainBERT on ADFD; and BrainWave on ADHD-Adult. These recurring gains across shot settings are unlikely to arise from a single support-set sample, suggesting that prototype transfer can effectively exploit the geometry in frozen representations for some model-dataset pairs.

Exploratory Structural Analyses

Q3: What structural patterns can be observed in BFM representations? To address this question, we examine spatial sensitivity, frequency-domain structure, and discrete representations. These analyses characterize structural tendencies rather than establishing causal learning mechanisms.

Spatial Sensitivity

Several high-performing BFMs in our benchmark, including BrainWave, BrainOmni, and REVE, explicitly model channel identities or cross-channel interactions, motivating an analysis of their sensitivity to spatial organization. We therefore jointly permute signal channels and their corresponding channel identifiers during training, preserving signal-channel correspondence, while evaluating on the canonical test split.

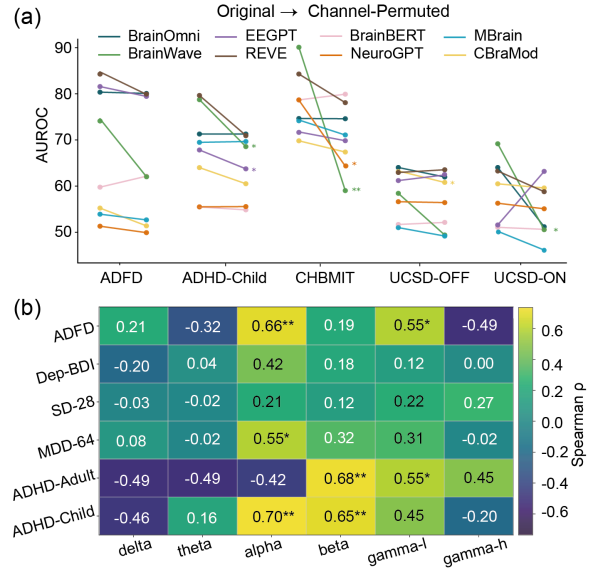


Figure 3: Structural Analyses. (a) AUROC changes under channel permutation. (b) Spearman correlation between band-wise predictability and model performance rankings. Asterisks indicate * ($p < 0.05$) and ** ($p < 0.01$).

We compare models with explicit spatial modeling (BrainWave, BrainOmni, CBraMod, EEGPT, MBrain, and REVE) with models that use less explicit spatial information (BrainBERT and NeuroGPT-E) by measuring changes in AUROC and applying paired significance tests (Figure 3(a)); complete results are reported in the supplementary material.

Performance changes under channel permutation vary across tasks, indicating different dependencies on spatial organization. We use spatial sensitivity to encompass both spatial heterogeneity in region-dependent channel statistics and topological sensitivity to relative channel organization. The perturbation preserves per-channel temporal statistics while changing the canonical input order and, for order-dependent spatial modules, the induced channel topology, thereby measuring the combined effects of these two factors. Epilepsy tasks show the largest AUROC drops, consistent with the importance of spatially distributed seizure onset and propagation patterns (Proix et al. 2020). PD exhibits intermediate sensitivity, in line with previously reported region-dependent spectral abnormalities (Barone and Rossiter 2021). ADHD shows smaller changes, suggesting weaker dependence on the presented channel organization under our evaluation setting, despite reported frontal-network alterations (Murias, Swanson, and Srinivasan 2007). Most models are also relatively stable on AD, where commonly used EEG biomarkers are dominated by broad spectral changes rather than narrowly localized activity patterns (Babiloni et al. 2004).

Robustness to channel permutation also differs across model architectures. BrainBERT, which does not explicitly model cross-channel interactions, is largely insensitive to the perturbation, whereas NeuroGPT-E degrades on several datasets. Among models with explicit spatial components, BrainWave, which uses CNNs to capture spatial information,

shows notable drops on CHBMIT, ADHD-Child, and ADFD, indicating sensitivity to the channel organization presented during training. EEGPT also remains sensitive on several spatially heterogeneous tasks despite its channel-alignment mechanism. By contrast, BrainOmni, CBraMod, MBrain, and REVE are generally more stable across datasets. These models incorporate graph structures, spatial embeddings, or semantic channel modeling to represent multichannel signals. However, robustness to channel reconfiguration is not sufficient: spatial modules that rely on fixed channel layouts remain difficult to transfer across iEEG patients with heterogeneous implantations. Learning structure-aware yet layout-flexible representations remains an open challenge.

Frequency-Domain Associations

Several BFM models that explicitly reconstruct frequency-domain features, including BrainWave, BrainBERT, and LaBraM, perform strongly on multiple tasks. This motivates us to examine whether their task-adapted representations retain task-relevant spectral information, given that neural oscillations across frequency bands are associated with distinct cognitive and pathological states (Buzsaki and Draguhn 2004). We freeze each fine-tuned encoder and train a lightweight PSD probe to predict the spectral profile of six frequency bands (Figure 3(b)). Separate linear heads map the frozen representation to the PSD of each band. Let $P_b(x)$ and $\hat{P}_b(x)$ denote the ground-truth and predicted PSD for band b . We quantify band predictability as $r_b = \text{corr}(P_b(x), \hat{P}_b(x))$ and normalize it across bands to obtain the relative predictive strength r_b^n . We then compute Spearman correlations between r_b^n and downstream model rankings, primarily based on AUROC. Complete results are reported in the supplementary material.

The associations are strongly task dependent. On ADFD, the relative predictability of α and low- γ is significantly associated with downstream ranking, whereas no band shows a significant association on the depression datasets. On ADHD-Adult, β and low- γ show significant positive correlations; on ADHD-Child, significant associations are observed for α and β . These patterns are broadly consistent with prior reports of frequency-specific biomarkers, including altered δ/α and θ/α ratios in AD (Newson and Thiagarajan 2019) and θ/β -related markers in ADHD (Loo et al. 2013). These results indicate that task-adapted BFM representations emphasize different frequency bands across tasks, some of which are associated with downstream performance.

Codebook Representations

Discrete codebooks have recently been adopted in several BFMs to compress continuous signals or latent features into reusable tokens. We analyze three models with explicit discretization (BFM, LaBraM, and BrainOmni), which differ in where and how quantization is applied. BFM uses the Chronos tokenizer to map raw signals into token IDs. LaBraM learns a VQ-VAE codebook over encoder features during pretraining but removes it during downstream fine-tuning. BrainOmni adopts a related design with multi-level residual vector quantization (RVQ). On the test set, we characterize token usage through entropy (Ent), and evalu-

	ADFD			CHBMIT			SZ-28		
Model	Ent	Inter	DR	Ent	Inter	DR	Ent	Inter	DR
BFM	.868	.998	.002	.838	.999	.001	.853	.999	.001
BrainOmni ₁	.947	.996	.007	.948	.999	.001	.941	.999	.002
BrainOmni ₂	.891	.928	.077	.950	.966	.037	.982	.994	.007
BrainOmni ₃	.929	.927	.076	.970	.959	.043	.973	.954	.049
BrainOmni ₄	.946	.947	.055	.988	.930	.072	.972	.909	.094
LaBraM _{no}	.755	.995	.006	.721	.973	.038	.607	.986	.022
LaBraM	.694	.267	1.155	.518	-.954	6.755	.586	.800	.294

Table 5: Codebook usage and class geometry across datasets. Ent, Inter, and DR denote entropy, mean inter-class similarity, and inter-/intra-class distance ratio. LaBraM_{no} uses the frozen codebook, and BrainOmni₁₋₄ denote 4 RVQ levels.

ate token-embedding geometry using class-center similarity (Inter) and the inter-to-intra-class distance ratio (DR).

The three tokenization schemes show distinct relationships between codebook utilization and class geometry (Table 44). BFM uses a moderate fraction of its vocabulary but distributes assignments relatively evenly among active tokens; however, its high Inter and low DR indicate weak class separation. LaBraM uses fewer and less evenly distributed tokens and shows stronger class separation than BFM, although its frozen codebook remains weakly discriminative without downstream adaptation. BrainOmni maintains near-complete utilization and high entropy across RVQ levels, with deeper levels generally showing larger DR, particularly on CHBMIT and SD-28. Retaining and updating LaBraM’s codebook during full fine-tuning further reduces coverage and entropy while increasing class separation. This variant outperforms frozen transfer but remains below standard full fine-tuning without quantization. By contrast, retaining BrainOmni’s codebook yields a modest but statistically non-significant improvement over its no-codebook variant. Overall, discrete tokenization is not uniformly beneficial for electrical brain signal decoding. Its effect varies across models and tasks and should be interpreted jointly with codebook utilization, class geometry, and downstream adaptation.

Conclusion

Brain4FMs provides a unified and extensible framework for evaluating BFMs on EEG/iEEG classification. We benchmark 17 models across 21 public datasets under standardized cross-subject protocols. Our goal is to assess how well existing BFMs transfer to EEG/iEEG analysis. Beyond task performance, we examine modality-dependent behavior, the effects of adaptation, and the representation structure. Together, these analyses characterize the current capabilities and limitations of BFMs and highlight the need for stronger generalization across subjects, devices, and tasks. The supplementary material provides detailed benchmark settings and complete experimental results. We currently focus on classification because it remains central to brain-signal analysis and is the primary downstream setting supported by existing BFMs. Generative tasks are not yet covered and will be incorporated as the field and available models mature.

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Appendix

Experimental Details

This section details the dataset preprocessing and experimental protocols used in our benchmark. We describe the unified evaluation setup adopted across models and tasks, together with the learning-rate search and model-compatible optimization settings used to ensure consistent and reproducible comparisons.

Dataset Details

Table 6 summarizes dataset characteristics omitted from the main text for brevity. We select datasets by balancing diversity, practical availability, and minimal dataset-level overlap with the pretraining corpora of the evaluated BFMs. These details reflect the dataset acquisition protocols and facilitate benchmark reproduction.

All datasets are processed under consistent cross-subject cross-validation protocols. Depending on the dataset and task protocol, preprocessing may include band-pass filtering, notch filtering, downsampling, segmentation into fixed-length non-overlapping samples or trials, removal of non-EEG/iEEG channels (e.g., EMG and EOG), and per-channel z-score normalization. Preprocessing is defined at the dataset level and shared by all models to avoid model-specific signal-processing bias. Chisco and DEAP use the officially preprocessed data. For the remaining datasets, we apply notch filtering at 50 or 60 Hz to remove power-line noise when applicable. The complete dataset-specific preprocessing configurations, including band-pass ranges, target sampling rates, and segment lengths, are reported in Table 6.

We do not impose an additional unified bipolar montage or re-referencing scheme. Instead, we preserve the channel montage and reference configuration released with each public dataset. For example, ISRUC-G1 retains its original contralateral-mastoid reference setting, while CHBMIT retains its original bipolar montage. Dataset-specific preprocessing procedures are implemented in our open-source code and documented in the `data_preprocess` directory. Since preprocessing and storage are performed independently within each subject or patient, no cross-subject information leakage is introduced. All datasets used in Brain4FMs are publicly released; ethics and fairness considerations are therefore limited, as no new data collection is involved.

Potential overlap between downstream datasets and pretraining corpora may introduce uncontrolled bias, which is a common challenge in evaluating released foundation models. We therefore minimized such overlap during downstream dataset selection whenever possible. Selected datasets are disjoint from the dominant reported pretraining sources, such as TUH (Obeid and Picone 2016) and TUEG (Harati et al. 2014). The treatment of known pretraining-downstream overlap is transparent. The manuscript identifies the potentially overlapping NeuroLM/LaBraM on SEED-IV and MVPFormer on SWEC.

To preserve benchmark completeness and provide a transparent account of all available evaluations, we report the performance of every evaluated model-dataset pair, including pairs with potential pretraining overlap. These potentially

affected entries are, however, excluded from all primary analyses and conclusions in the main text. Specifically, we exclude NeuroLM and LaBraM on SEED-IV, for which exposure to related emotion data cannot be ruled out, as well as MVPFormer on SWEC, since SWEC was used as part of its pretraining data. These results are retained in the appendix solely for coverage and transparency. Removing the potentially overlapping pairs may change individual scores and some aggregate summary values, but it does not materially alter the qualitative conclusions drawn from the main-text analyses.

Experimental Setup

We evaluate all models under a unified cross-subject cross-validation protocol. For each fold, subjects in the training, validation, and test sets are kept disjoint to prevent subject-level information leakage. Models are fine-tuned for up to 50 epochs with early stopping using a patience of 5. We use Adam-family optimizers, i.e., Adam or AdamW following the official implementations (7), with a batch size of 128 and no learning rate scheduler.

For downstream adaptation, we retain each BFM’s released input transformation, representation aggregation, classification interface, and compatible Adam-family optimizer, while applying a standardized training and hyperparameter-selection protocol. Specifically, data splits, preprocessing, learning-rate search, training budget, checkpoint selection, and evaluation metrics are standardized across models. This design preserves model-compatible readout mechanisms without reproducing model-specific fine-tuning recipes.

For transparency, all classification head types, head-parameter formulas, and optimizer choices are reported in Table 7. The number of trainable head parameters is computed after instantiating each model under the corresponding downstream configuration. It is task- and configuration-dependent, varying with the number of target classes C , the final representation dimension, and, for models whose flattened feature dimension depends on temporal tokenization, the sequence length S and the number of channels N_{ch} . Accordingly, Table 7 reports the symbolic parameter count of the downstream classification head.

To facilitate exact reproduction, we release fixed subject-level splits, dataset-specific preprocessing scripts, model wrappers, pretrained weight version records, and all training and evaluation configuration files in the code repository. These artifacts ensure that all reported results can be reproduced under the same data splits, preprocessing pipeline, and fine-tuning protocol.

Learning Rate Search

To examine the sensitivity of full-FT to learning rate (LR), we conduct a focused LR search on 5 representative BFMs spanning different model scales. These include the lightweight SppEEGNet (138K), medium-scale models LaBraM (5.80M) and MBrain (8.34M), and large-scale models BrainWave (102.13M) and Brant (196.10M). Figure 4 reports their AUROC under five candidate encoder lrs, i.e., $\{1 \times 10^{-4}, 5 \times 10^{-5}, 1 \times 10^{-5}, 5 \times 10^{-6}, 1 \times 10^{-6}\}$, on 5 downstream datasets: ADFD, CHBMIT, FNUSA, SZ-28,

Name	Signal	Task	Subject	#Ch	SeqLen	sfreq (Hz)	Passband (Hz)	Total Time	Fold	Cat.
CHBMIT	EEG	Epilepsy	23 sub.	23	10s	256	0.01-256/3	686 h	5	2
MAYO	iEEG	DRE	25 sub.	1	3s	5000 $\rightarrow$ 1000	0.01-1000/3	94 h	6	2
FNUSA	iEEG	DRE	14 sub.	1	3s	5000 $\rightarrow$ 1000	0.01-1000/3	150 h	5	2
HUP-EECoG	iEEG	DRE	20 sub.	54-125	2s	1024 $\rightarrow$ 512	0.5-120	1020 min	5	2
HUP-SEEG	iEEG	DRE	38 sub.	36-195	2s	1024 $\rightarrow$ 512	0.5-120	38 h	5	2
SWEC	iEEG	DRE	68 sub.	22-128	5s	512/1024 $\rightarrow$ 512	0.5-150	9328 h	5	2
Dep-BDI	EEG	Depression	122 sub.	64	10s	500 $\rightarrow$ 250	0.01-250/3	32 h	5	2
Dep-STAI	EEG	Anxiety	122 sub.	64	10s	500 $\rightarrow$ 250	0.01-250/3	32 h	5	3
MDD-64	EEG	MDD	30H, 43MDD	19	10s	256	0.5-70	21 h	5	2
SZ-28	EEG	SZ	28 sub.	19	5s	250	0.2-250/3	958 min	5	2
UCSD-ON	EEG	PD	31H, 15PD	32	10s	512 $\rightarrow$ 256	0.01-256/3	220 min	5	2
UCSD-OFF	EEG	PD	31H, 15PD	32	10s	512 $\rightarrow$ 256	0.01-256/3	220 min	5	2
ADFD	EEG	AD	88 sub.	19	10s	500 $\rightarrow$ 250	0.01-250/3	1164 min	5	2
ADHD-Adult	EEG	ADHD	42H, 37ADHD	2	5s	256	0.01-256/3	423 min	5	2
ADHD-Child	EEG	ADHD	60H, 61ADHD	19	5s	128	0.5-128/3	275 min	5	2
ISRUC-G1	EEG	Sleep Stage	100 sub.	6	30s	200	0.3-45	745 h	5	5
SleepEDF	EEG	Sleep Stage	44 sub.	1	30s	100	0.3-45	378.7 h	5	5
EEGMMIDB-R	EEG	ME	109 sub.	64	4s	160	0.01-70	2834 min	6	4
EEGMMIDB-I	EEG	MI	109 sub.	64	4s	160	0.01-70	2834 min	6	4
BCI-2a	EEG	MI	9 sub.	22	3s	250	0.5-100	130 min	4	4
Chisco-R	EEG	Concept Classification	5 sub.	125	3.3s	1000 $\rightarrow$ 500	1-Nyquist (<250)	59 h	5	39
Chisco-I	EEG	Concept Classification	5 sub.	125	3.3s	1000 $\rightarrow$ 500	1-Nyquist (<250)	59 h	5	39
Cogitate-CF	iEEG	Visual Classification	19 sub.	6-142	2s	2048 $\rightarrow$ 512	0.1-150	120 min	5	2
DEAP	EEG	Emotion Recognition	32 sub.	32	10s	128	4-45	2133 min	5	4
SEED-IV	EEG	Emotion Recognition	15 sub.	62	4s	200	1-75	54 h	6	4
EEGMat	EEG	MW	36 sub.	32	10s	500 $\rightarrow$ 250	0.1-75	144 min	4	2

Table 6: Public EEG and iEEG datasets used in the benchmark, including signal type, task, subjects (Sub.), number of channels (#Ch), sequence length (SeqLen), sampling frequency (sfreq), effective preprocessing passband (Passband), total recording duration, cross-validation folds (Fold) and number of categories (Cat.); H denotes healthy subjects.

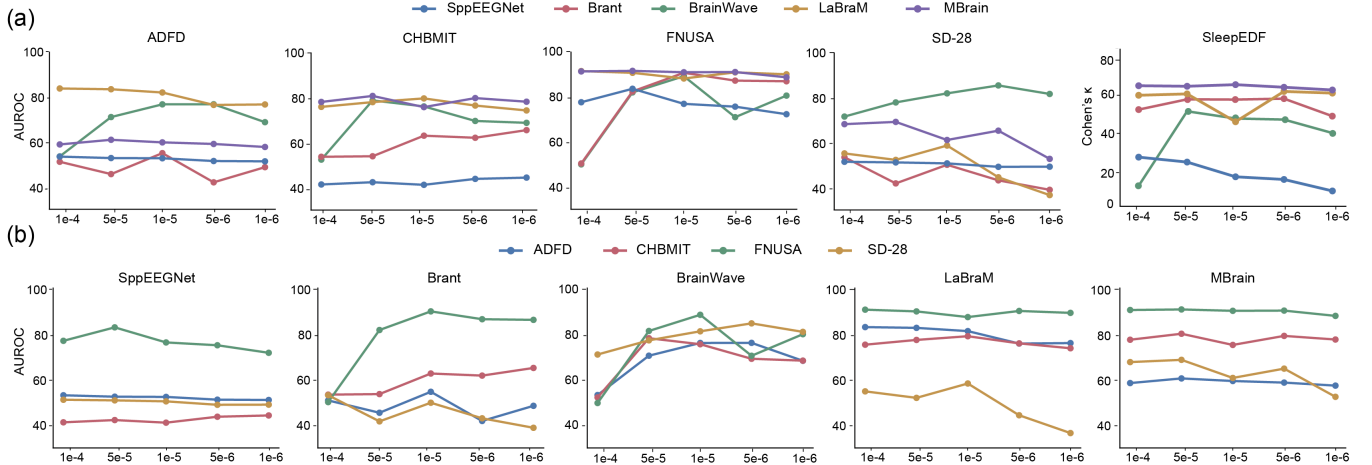


Figure 4: Learning rate search for full-FT across datasets and models. (a) Results grouped by dataset, enabling direct comparison of different models on each dataset. (b) Results grouped by model, highlighting dataset-specific LR sensitivity for each architecture.

and SleepEDF. For all experiments, the downstream head LR is set to $10\times$ the encoder LR.

Figure 4 reports the task-specific primary validation metric under different encoder lrs. Following the main evaluation protocol, AUROC is used for binary classification datasets,

while Cohen's κ is used for the multi-class SleepEDF dataset. The results show that model performance is highly sensitive to the choice of LR. First, the optimal LR differs across models even on the same dataset. For example, on CHBMIT, Brant tends to favor smaller lrs, whereas SppEEGNet

achieves better performance with relatively larger lrs. This indicates that different model architectures can exhibit distinct fine-tuning dynamics, even when evaluated on the same downstream task. The optimal LR for the same model also varies across datasets. For instance, Brant achieves its best performance with an encoder LR of 1×10^{-5} on FNUSA, while its optimal encoder LR shifts to 1×10^{-6} on CHBMIT. This dataset-dependent behavior suggests that the fine-tuning stability of a given model is affected not only by the model architecture, but also by the characteristics of the downstream dataset.

These observations demonstrate that using a fixed LR for all models and datasets would introduce model- and dataset-dependent bias. Therefore, we perform LR search independently for each model-dataset pair. For each fold, we select the checkpoint with the lowest validation loss for evaluation. The best encoder LR is chosen according to the average task-specific primary validation metric across folds, i.e., AUROC for binary tasks and Cohen’s κ for multi-class tasks. Final performance is reported as the average test performance across folds using the selected encoder LR and its corresponding head LR. The selected encoder LR for each model-dataset pair used in full-FT is summarized in Table 8.

Additional Experimental Results

This section reports the complete results underlying the main findings reported in the paper. We use full-parameter fine-tuning as the primary setting because Brain4FMs evaluates the downstream performance attainable after adapting each pretrained model to the target task, whereas frozen-encoder transfer serves as a complementary measure of fixed-representation transferability. Empirically, encoder adaptation improves AUROC by 3.26 points on average, with particularly large gains for some model-dataset pairs. It includes the full benchmark results under full-parameter fine-tuning, detailed frozen-encoder and prototype-based transfer

results, and complete exploratory analyses of spatial sensitivity, frequency-domain associations, and codebook representations.

Complete Benchmark Results

We report the complete evaluation results of all 17 BFM across 26 downstream classification settings. All 17 BFMs are evaluated on 22 settings, while four heterogeneous multi-channel iEEG settings support 10 compatible models (BIOT, LaBraM, CBraMod, BrainBERT, SppEEGNet, BrainWave, MBrain, BFM, CodeBrain and MVPFormer), yielding 414 model-setting pairs. For NeuroGPT, we additionally report both encoder-only (NeuroGPT-E) and encoder-decoder (NeuroGPT-D) variants on the 22 applicable settings, resulting in 436 reported pairs overall. For binary tasks, we report AUROC, Accuracy, F1, and F2 to capture threshold-free discrimination and positive-class detection under imbalance. For multi-class tasks, we report Accuracy, AUROC (one-vs-rest; OvR), macro-F1 (MF1), and Cohen’s κ (Kappa) to measure overall discrimination, class-wise performance, and agreement beyond chance. We report results as mean $\pm$ standard deviation over n-fold cross-validation, and bold indicates the highest mean performance based on unrounded values. All metrics use two decimal places, except Cohen’s κ , which uses three decimals due to its small magnitude on some datasets. In Tables 9 to 34, models are grouped by SSL paradigm and ranked by AUROC within each group.

Model	Downstream Head	Head Params	Optimizer
BrainBERT	ReLU+Linear	$295296 + 385C$	AdamW
Brant	ReLU+Linear	$524800 + 513C$	AdamW
SppEEGNet	ReLU+Linear	$5050 + 51C$	Adam
BFM	ReLU+Linear	$524800 + 513C$	AdamW
CBraMod	ReLU+Linear	$20000S^2 + 100S(C + 1) + C$	AdamW
NeuroLM	ReLU+Linear	$295296 + 385C$	AdamW
LaBraM	ReLU+Linear	$201C$	AdamW
Bendr	Norm+Linear	$513C$	Adam
MBrain	Linear	$229760 + 129C$	Adam
NeuroGPT	Linear	$1320224 + 33C$	AdamW
EEGPT	Linear	$1320224 + 33C$	AdamW
BrainOmni	Linear	$262144N_{\text{ch}} + 512 + 513C$	AdamW
BrainWave	CNN	$295296 + 385C$	AdamW
BIOT	ELU+Linear	$257C$	Adam
REVE	Norm+Linear	$512(\lfloor \frac{200(S-1)}{180} \rfloor + 1)N_{\text{ch}}(C + 1) + C$	AdamW
MVPFormer	ReLU+Linear	$295296 + 385C$	AdamW
CodeBrain	ReLU+Linear	$20100 + 101C$	AdamW

Table 7: Downstream classification interfaces, trainable head-parameter formulas, and optimizers. The reported formulas count trainable parameters in the downstream classification head only, excluding the pretrained encoder. C denotes the number of target classes, S the effective sequence length, and N_{ch} the number of input channels.

Dataset	BIOT	Bendr	BrainBERT	BrainOmni	BrainWave	Brant	CBraMod	EEGPT	LaBraM	MBrain	NeuroGPT	NeuroLM	REVE	SppEEGNet	BFM	MVPFormer	CodeBrain
MAYO	10^{-4}	10^{-4}	5×10^{-5}	5×10^{-5}	5×10^{-5}	5×10^{-5}	10^{-4}	5×10^{-5}	5×10^{-5}	5×10^{-5}	5×10^{-5}	5×10^{-5}	10^{-5}	10^{-4}	10^{-5}	5×10^{-5}	10^{-5}
SEED-IV	5×10^{-5}	5×10^{-5}	10^{-5}	5×10^{-5}	10^{-5}	10^{-5}	5×10^{-5}	10^{-5}	5×10^{-5}	5×10^{-5}	10^{-5}	10^{-5}	10^{-5}	5×10^{-5}	10^{-5}	10^{-5}	5×10^{-5}
EEGMMIDB-I	5×10^{-5}	5×10^{-5}	10^{-5}	5×10^{-5}	10^{-5}	10^{-5}	5×10^{-5}	10^{-5}	5×10^{-5}	5×10^{-5}	10^{-5}	10^{-5}	10^{-5}	5×10^{-5}	10^{-5}	10^{-5}	5×10^{-5}
EEGMMIDB-R	5×10^{-5}	5×10^{-5}	10^{-5}	5×10^{-5}	10^{-5}	10^{-5}	5×10^{-5}	10^{-5}	5×10^{-5}	5×10^{-5}	10^{-5}	10^{-5}	10^{-5}	5×10^{-5}	10^{-5}	10^{-5}	5×10^{-5}
UCSD-ON	5×10^{-5}	5×10^{-5}	10^{-5}	10^{-5}	10^{-5}	10^{-5}	5×10^{-5}	10^{-5}	10^{-5}	10^{-5}	10^{-5}	10^{-5}	10^{-5}	5×10^{-5}	10^{-5}	10^{-5}	10^{-5}
Dep-STAI	5×10^{-5}	5×10^{-5}	10^{-5}	5×10^{-5}	10^{-5}	10^{-5}	5×10^{-5}	10^{-5}	5×10^{-5}	5×10^{-5}	10^{-5}	10^{-5}	10^{-5}	5×10^{-5}	10^{-5}	10^{-5}	5×10^{-5}
FNUSA	10^{-4}	10^{-4}	5×10^{-5}	5×10^{-5}	10^{-5}	10^{-5}	10^{-4}	5×10^{-5}	10^{-4}	5×10^{-5}	10^{-4}	5×10^{-5}	5×10^{-6}	10^{-4}	10^{-5}	5×10^{-5}	5×10^{-5}
UCSD-OFF	5×10^{-5}	5×10^{-5}	10^{-5}	10^{-5}	10^{-5}	10^{-5}	5×10^{-5}	10^{-5}	10^{-5}	10^{-5}	10^{-5}	10^{-5}	10^{-5}	5×10^{-5}	10^{-5}	10^{-5}	10^{-5}
Dep-BDI	10^{-4}	10^{-4}	5×10^{-5}	5×10^{-5}	5×10^{-5}	5×10^{-5}	10^{-4}	5×10^{-5}	5×10^{-5}	5×10^{-5}	5×10^{-5}	5×10^{-5}	10^{-5}	10^{-4}	10^{-5}	5×10^{-5}	5×10^{-5}
DEAP	5×10^{-5}	5×10^{-5}	10^{-5}	5×10^{-5}	10^{-5}	10^{-5}	5×10^{-5}	10^{-5}	5×10^{-5}	5×10^{-5}	10^{-5}	10^{-5}	10^{-5}	5×10^{-5}	5×10^{-5}	10^{-5}	5×10^{-5}
Chisco-R	5×10^{-5}	5×10^{-5}	10^{-5}	10^{-5}	10^{-5}	10^{-5}	5×10^{-5}	10^{-5}	10^{-5}	10^{-5}	10^{-5}	10^{-5}	10^{-5}	5×10^{-5}	10^{-5}	10^{-5}	10^{-5}
Chisco-I	5×10^{-5}	5×10^{-5}	10^{-5}	10^{-5}	10^{-5}	10^{-5}	5×10^{-5}	10^{-5}	10^{-5}	10^{-5}	10^{-5}	10^{-5}	10^{-5}	5×10^{-5}	10^{-5}	10^{-5}	10^{-5}
CHBMIT	5×10^{-5}	5×10^{-5}	10^{-5}	10^{-5}	5×10^{-5}	10^{-6}	10^{-4}	10^{-5}	10^{-5}	5×10^{-5}	10^{-5}	10^{-5}	5×10^{-5}	10^{-4}	10^{-6}	10^{-5}	5×10^{-5}
ADHD-Child	5×10^{-5}	5×10^{-5}	10^{-5}	5×10^{-5}	10^{-5}	10^{-5}	5×10^{-5}	10^{-5}	5×10^{-5}	5×10^{-5}	10^{-5}	10^{-5}	10^{-5}	5×10^{-5}	10^{-5}	10^{-5}	5×10^{-5}
ADFD	10^{-4}	10^{-4}	5×10^{-5}	5×10^{-5}	5×10^{-6}	5×10^{-5}	10^{-4}	5×10^{-5}	10^{-4}	5×10^{-5}	5×10^{-5}	5×10^{-5}	10^{-5}	10^{-4}	10^{-5}	10^{-5}	10^{-5}
ADHD-Adult	5×10^{-5}	5×10^{-5}	10^{-5}	5×10^{-5}	10^{-5}	10^{-5}	5×10^{-5}	10^{-5}	5×10^{-5}	5×10^{-5}	10^{-5}	10^{-5}	10^{-5}	5×10^{-5}	10^{-5}	10^{-5}	5×10^{-5}
MDD-64	10^{-4}	10^{-4}	10^{-5}	5×10^{-5}	10^{-5}	10^{-5}	5×10^{-6}	10^{-5}	10^{-5}	10^{-5}	10^{-5}	10^{-5}	5×10^{-6}	10^{-4}	10^{-5}	10^{-4}	5×10^{-5}
SZ-28	5×10^{-5}	5×10^{-5}	10^{-5}	5×10^{-5}	5×10^{-6}	10^{-5}	5×10^{-5}	10^{-5}	10^{-5}	5×10^{-5}	10^{-5}	10^{-5}	10^{-5}	10^{-4}	5×10^{-6}	5×10^{-5}	5×10^{-5}
SleepEDF	5×10^{-5}	5×10^{-5}	10^{-5}	5×10^{-5}	5×10^{-5}	10^{-5}	5×10^{-5}	10^{-5}	5×10^{-5}	10^{-4}	10^{-5}	10^{-5}	10^{-5}	10^{-4}	10^{-5}	10^{-4}	10^{-4}
ISRUC	5×10^{-5}	5×10^{-5}	10^{-5}	5×10^{-5}	10^{-5}	10^{-5}	5×10^{-5}	10^{-5}	5×10^{-5}	5×10^{-5}	10^{-5}	10^{-5}	10^{-5}	5×10^{-5}	10^{-5}	5×10^{-5}	10^{-4}
BCI-2a	5×10^{-5}	5×10^{-5}	10^{-5}	10^{-5}	10^{-5}	10^{-5}	5×10^{-5}	10^{-5}	10^{-5}	10^{-5}	10^{-5}	10^{-5}	10^{-5}	5×10^{-5}	10^{-5}	10^{-5}	5×10^{-5}
EEGMat	5×10^{-5}	5×10^{-5}	10^{-5}	10^{-5}	10^{-5}	10^{-5}	5×10^{-5}	10^{-5}	10^{-5}	10^{-5}	10^{-5}	10^{-5}	10^{-5}	5×10^{-5}	10^{-5}	10^{-5}	5×10^{-5}
HUP-ECOG	10^{-4}	-	5×10^{-5}	-	5×10^{-5}	-	10^{-4}	-	5×10^{-5}	5×10^{-5}	-	-	-	10^{-4}	10^{-5}	10^{-5}	10^{-5}
HUP-SEEG	10^{-4}	-	5×10^{-5}	-	5×10^{-5}	-	10^{-4}	-	5×10^{-5}	5×10^{-5}	-	-	-	10^{-4}	10^{-5}	5×10^{-5}	10^{-5}
Cogitate-CF	5×10^{-5}	-	5×10^{-5}	-	10^{-5}	-	10^{-4}	-	5×10^{-5}	5×10^{-5}	-	-	-	10^{-4}	5×10^{-5}	10^{-5}	5×10^{-5}
SWEC	10^{-4}	-	5×10^{-5}	-	5×10^{-5}	-	10^{-4}	-	5×10^{-5}	5×10^{-5}	-	-	-	10^{-4}	10^{-5}	10^{-5}	10^{-5}

Table 8: Best learning rates selected for each dataset-model pair in our benchmark.

Type	Model	AUROC	Acc	F1	F2
C	MBrain	.75±.08	.71±.12	.54±.10	.60±.09
	BIOT	.56±.08	.29±.05	.41±.03	.61±.02
	Bendr	.55±.03	.59±.03	.32±.02	.36±.04
	SppEEGNet	.43±.05	.60±.06	.31±.04	.33±.05
G	BrainWave	.90±.03	.80±.12	.71±.09	.78±.04
	REVE	.84±.10	.78±.12	.63±.14	.66±.13
	NeuroGPT-E	.78±.12	.75±.07	.50±.12	.63±.16
	BrainBERT	.78±.06	.75±.10	.58±.08	.61±.09
	BrainOmni	.75±.11	.71±.12	.51±.12	.54±.12
	LaBraM	.75±.10	.73±.10	.53±.14	.57±.17
	BFM	.74±.05	.72±.03	.51±.09	.51±.14
	CBraMod	.69±.04	.70±.08	.46±.08	.49±.14
	NeuroGPT-D	.61±.07	.73±.03	.18±.23	.19±.25
	CodeBrain	.60±.08	.74±.10	.52±.09	.52±.11
	Brant	.60±.09	.72±.06	.08±.12	.07±.11
O	EEGPT	.71±.09	.67±.11	.46±.06	.51±.15
	NeuroLM	.67±.06	.54±.20	.27±.23	.38±.34
	MVPFormer	.55±.02	.70±.02	.29±.03	.26±.04

Table 9: Performance of BFM on the CHBMIT.

Type	Model	AUROC	Acc	F1	F2
C	Bendr	.93±.03	.90±.02	.69±.13	.75±.11
	MBrain	.92±.04	.92±.02	.70±.11	.73±.09
	BIOT	.90±.07	.88±.05	.63±.14	.72±.13
	SppEEGNet	.80±.03	.86±.05	.57±.14	.60±.10
G	BrainWave	.98±.01	.93±.02	.81±.09	.86±.03
	BrainBERT	.97±.01	.95±.01	.81±.07	.83±.07
	LaBraM	.96±.02	.93±.02	.76±.12	.80±.09
	NeuroGPT-E	.96±.02	.93±.02	.71±.07	.67±.08
	REVE	.92±.04	.89±.04	.66±.13	.72±.11
	Brant	.92±.03	.82±.12	.58±.19	.69±.14
	BrainOmni	.91±.06	.91±.02	.67±.13	.69±.12
	CBraMod	.89±.04	.87±.02	.59±.12	.64±.10
	CodeBrain	.89±.02	.91±.03	.74±.11	.76±.11
	BFM	.81±.08	.68±.08	.43±.15	.59±.14
O	NeuroGPT-D	.77±.15	.88±.03	.23±.26	.19±.21
	EEGPT	.92±.03	.90±.03	.67±.10	.70±.07
	MVPFormer	.68±.08	.72±.06	.34±.07	.39±.10
	NeuroLM	.64±.09	.72±.16	.30±.20	.37±.26

Table 10: Performance of BFM on the MAYO.

Type	Model	AUROC	Acc	F1	F2
C	MBrain	.91±.08	.87±.08	.75±.13	.76±.16
	Bendr	.88±.06	.84±.06	.73±.11	.77±.10
	BIOT	.87±.07	.83±.08	.73±.14	.77±.10
	SppEEGNet	.82±.09	.84±.09	.73±.09	.73±.09
G	BrainWave	.92±.05	.89±.06	.83±.06	.83±.04
	NeuroGPT-E	.90±.06	.86±.05	.74±.04	.72±.11
	BrainBERT	.90±.04	.89±.04	.77±.09	.79±.08
	LaBraM	.89±.08	.82±.10	.72±.15	.76±.11
	BrainOmni	.88±.07	.84±.05	.73±.08	.75±.09
	REVE	.87±.08	.82±.07	.71±.10	.74±.08
	Brant	.87±.12	.84±.07	.75±.06	.79±.08
	CodeBrain	.83±.10	.85±.06	.78±.06	.76±.09
	BFM	.78±.12	.69±.11	.62±.05	.69±.08
	NeuroGPT-D	.72±.12	.77±.12	.30±.38	.27±.35
	CBraMod	.71±.13	.78±.08	.62±.13	.64±.12
O	EEGPT	.90±.04	.84±.06	.75±.11	.78±.08
	NeuroLM	.64±.14	.72±.17	.37±.28	.38±.35
	MVPFormer	.59±.07	.63±.06	.37±.07	.38±.11

Table 11: Performance of BFM on the FNUSA.

Type	Model	AUROC	Acc	F1	F2
C	MBrain	.57±.09	.61±.12	.02±.05	.02±.04
	BIOT	.54±.08	.61±.12	.19±.13	.40±.06
	SppEEGNet	.44±.07	.52±.09	.35±.10	.36±.10
G	BrainWave	.88±.07	.80±.11	.72±.13	.65±.18
	BrainBERT	.81±.12	.73±.09	.63±.11	.60±.23
	BFM	.72±.11	.71±.10	.47±.13	.41±.18
	CodeBrain	.67±.05	.71±.06	.33±.16	.30±.20
	CBraMod	.64±.09	.58±.10	.39±.24	.45±.31
	LaBraM	.51±.06	.64±.13	.00±.00	.00±.00
O	MVPFormer	.59±.08	.62±.07	.46±.12	.44±.22

Table 12: Performance of BFM on the HUP-ECOG.

Type	Model	AUROC	Acc	F1	F2
C	SppEEGNet	.52±.02	.52±.19	.22±.16	.25±.24
	MBrain	.50±.08	.62±.15	.08±.18	.13±.28
	BIOT	.45±.08	.68±.21	.28±.39	.30±.41
G	BrainWave	.79±.11	.74±.14	.50±.20	.45±.25
	BFM	.61±.02	.74±.01	.19±.08	.13±.06
	CodeBrain	.61±.04	.69±.03	.29±.10	.25±.12
	BrainBERT	.57±.09	.67±.14	.00±.00	.00±.00
	CBraMod	.55±.08	.63±.17	.38±.15	.32±.13
	LaBraM	.53±.09	.63±.19	.34±.19	.30±.17
O	MVPFormer	.54±.04	.60±.18	.27±.09	.26±.20

Table 13: Performance of BFM on the HUP-SEEG.

Type	Model	AUROC	Acc	F1	F2
C	BIOT	.66±.06	.45±.19	.41±.01	.56±.10
	MBrain	.63±.05	.75±.00	.00±.00	.00±.00
	SppEEGNet	.44±.05	.57±.10	.34±.14	.42±.21
G	BrainWave	.84±.08	.82±.08	.61±.17	.58±.16
	LaBraM	.83±.10	.80±.07	.59±.17	.60±.21
	CBraMod	.78±.11	.75±.09	.62±.05	.70±.06
	BrainBERT	.77±.08	.61±.18	.48±.07	.54±.16
	BFM	.65±.10	.73±.04	.36±.21	.37±.24
	CodeBrain	.59±.07	.60±.17	.44±.07	.51±.11
O	MVPFormer	.62±.05	.64±.10	.43±.06	.49±.11

Table 14: Performance of BFM on the SWEC.

Type	Model	AUROC	Acc	F1	F2
C	BIOT	.56±.17	.49±.12	.58±.14	.70±.24
	MBrain	.53±.12	.50±.10	.53±.12	.52±.14
	Bendr	.52±.02	.52±.01	.54±.05	.52±.07
	SppEEGNet	.52±.02	.51±.02	.51±.05	.49±.07
G	REVE	.84±.07	.77±.08	.79±.07	.79±.09
	LaBraM	.83±.10	.75±.07	.77±.08	.76±.09
	BrainOmni	.80±.07	.71±.04	.73±.05	.72±.09
	BrainWave	.74±.05	.68±.05	.68±.10	.73±.10
	BrainBERT	.59±.05	.58±.05	.60±.11	.64±.15
	BFM	.59±.08	.58±.09	.59±.12	.61±.09
	CBraMod	.55±.06	.59±.04	.41±.20	.37±.20
	NeuroGPT-E	.51±.03	.55±.01	.71±.01	.85±.01
	Brant	.50±.07	.54±.04	.70±.03	.84±.04
	CodeBrain	.50±.09	.69±.05	.72±.04	.72±.04
	NeuroGPT-D	.50±.03	.55±.01	.71±.01	.86±.01
O	EEGPT	.81±.05	.72±.05	.73±.05	.70±.07
	MVPFormer	.52±.01	.53±.03	.56±.05	.55±.06
	NeuroLM	.51±.04	.53±.03	.63±.09	.71±.16

Table 15: Performance of BFM on the ADFD.

Type	Model	AUROC	Acc	F1	F2
C	BIOT	.96±.02	.90±.04	.89±.04	.90±.05
	MBrain	.95±.03	.92±.04	.91±.04	.91±.06
	Bendr	.62±.03	.61±.04	.60±.05	.61±.06
	SppEEGNet	.51±.04	.54±.06	.36±.30	.36±.35
G	BrainOmni	.96±.02	.91±.02	.90±.03	.89±.04
	BrainWave	.96±.03	.91±.04	.91±.04	.91±.05
	REVE	.96±.02	.91±.02	.91±.02	.90±.05
	Brant	.95±.03	.89±.04	.89±.04	.90±.05
	LaBraM	.95±.04	.91±.04	.92±.04	.91±.04
	CBraMod	.95±.02	.92±.04	.91±.04	.91±.07
	NeuroGPT-E	.91±.05	.84±.06	.83±.05	.82±.03
	CodeBrain	.88±.08	.92±.03	.92±.03	.92±.04
	BFM	.88±.04	.82±.04	.80±.04	.81±.05
	NeuroGPT-D	.88±.05	.82±.05	.82±.04	.85±.03
	BrainBERT	.74±.05	.69±.05	.61±.11	.57±.14
O	EEGPT	.96±.03	.91±.04	.90±.05	.89±.08
	NeuroLM	.82±.14	.75±.07	.74±.06	.77±.11
	MVPFormer	.50±.01	.51±.05	.51±.03	.53±.02

Table 16: Performance of BFM on the ADHD-Adult.

Type	Model	AUROC	Acc	F1	F2
C	BIOT	.75±.07	.67±.04	.76±.03	.86±.04
	MBrain	.69±.11	.63±.08	.66±.06	.66±.04
	SppEEGNet	.53±.05	.49±.04	.48±.07	.45±.08
	Bendr	.53±.02	.51±.01	.52±.02	.49±.02
G	REVE	.79±.07	.71±.04	.74±.06	.74±.08
	BrainWave	.78±.04	.71±.04	.74±.06	.74±.12
	BrainOmni	.71±.09	.62±.07	.63±.09	.61±.14
	BFM	.69±.11	.69±.09	.75±.07	.75±.04
	Brant	.69±.10	.58±.09	.51±.14	.44±.15
	LaBraM	.66±.12	.64±.08	.74±.06	.70±.07
	CodeBrain	.64±.09	.63±.06	.69±.07	.74±.11
	CBraMod	.64±.05	.64±.06	.73±.03	.81±.05
	NeuroGPT-E	.55±.02	.56±.03	.72±.02	.86±.01
	BrainBERT	.55±.03	.56±.03	.56±.07	.53±.09
	NeuroGPT-D	.47±.12	.56±.06	.64±.11	.69±.18
O	EEGPT	.67±.13	.64±.10	.68±.10	.68±.12
	NeuroLM	.64±.06	.60±.05	.71±.04	.80±.05
	MVPFormer	.54±.02	.57±.03	.61±.02	.60±.02

Table 17: Performance of BFM on the ADHD-Child.

Type	Model	AUROC	Acc	F1	F2
C	SppEEGNet	.53±.07	.51±.06	.59±.15	.71±.25
	Bendr	.52±.04	.51±.03	.48±.05	.47±.05
	BIOT	.51±.19	.49±.05	.56±.27	.69±.35
	MBrain	.50±.04	.44±.04	.32±.23	.34±.29
G	BrainOmni	.64±.09	.57±.06	.54±.16	.57±.26
	CBraMod	.63±.05	.58±.04	.46±.09	.40±.12
	REVE	.62±.19	.58±.17	.58±.19	.60±.24
	LaBraM	.59±.05	.56±.03	.57±.08	.60±.15
	BrainWave	.58±.06	.53±.10	.62±.12	.71±.19
	NeuroGPT-E	.56±.06	.54±.05	.57±.12	.62±.22
	BFM	.53±.03	.49±.12	.36±.09	.47±.17
	Brant	.52±.10	.45±.22	.42±.29	.58±.35
	BrainBERT	.51±.06	.54±.06	.49±.17	.49±.20
	CodeBrain	.51±.08	.54±.07	.55±.14	.57±.21
	NeuroGPT-D	.45±.07	.51±.05	.01±.01	.00±.01
O	EEGPT	.61±.13	.55±.08	.56±.10	.58±.17
	NeuroLM	.51±.07	.50±.05	.53±.15	.61±.23
	MVPFormer	.49±.04	.53±.07	.53±.09	.54±.10

Table 18: Performance of BFM on the UCSD-OFF.

Type	Model	AUROC	Acc	F1	F2
C	Bendr	.51±.05	.49±.05	.43±.05	.41±.06
	SppEEGNet	.49±.08	.50±.11	.46±.23	.48±.27
	MBrain	.49±.15	.46±.14	.37±.17	.34±.18
	BIOT	.46±.18	.49±.05	.65±.05	.81±.04
G	BrainWave	.69±.10	.53±.08	.60±.06	.69±.14
	REVE	.63±.19	.56±.13	.45±.29	.46±.33
	CodeBrain	.62±.11	.56±.10	.60±.11	.63±.14
	CBraMod	.60±.12	.49±.08	.18±.08	.13±.07
	NeuroGPT-E	.56±.10	.53±.07	.58±.10	.64±.13
	Brant	.56±.11	.52±.11	.29±.28	.31±.34
	BrainOmni	.53±.14	.49±.12	.49±.16	.51±.20
	LaBraM	.51±.22	.45±.11	.33±.19	.31±.19
	BrainBERT	.51±.05	.51±.06	.51±.11	.54±.17
	NeuroGPT-D	.46±.10	.47±.06	.29±.25	.31±.28
	BFM	.44±.08	.48±.07	.42±.17	.42±.20
O	MVPFormer	.58±.07	.57±.05	.50±.05	.46±.05
	EEGPT	.51±.21	.48±.17	.49±.15	.49±.14
	NeuroLM	.48±.13	.51±.18	.47±.24	.48±.29

Table 19: Performance of BFM on the UCSD-ON.

Type	Model	AUROC	Acc	F1	F2
C	MBrain	.68±.06	.75±.02	.13±.15	.10±.11
	BIOT	.59±.08	.27±.01	.42±.01	.65±.01
	Bendr	.54±.02	.66±.05	.34±.07	.34±.06
	SppEEGNet	.52±.03	.50±.07	.37±.04	.47±.09
G	REVE	.78±.09	.75±.08	.57±.10	.66±.10
	LaBraM	.71±.03	.68±.04	.48±.07	.53±.13
	NeuroGPT-E	.70±.04	.73±.04	.25±.20	.22±.20
	BrainOmni	.68±.07	.60±.07	.48±.04	.60±.06
	BrainWave	.66±.06	.51±.20	.44±.07	.55±.08
	CBraMod	.63±.06	.72±.04	.19±.18	.16±.17
	BrainBERT	.61±.07	.67±.06	.45±.09	.50±.13
	BFM	.60±.03	.66±.04	.35±.06	.36±.10
	CodeBrain	.59±.05	.62±.09	.49±.04	.58±.07
	Brant	.57±.12	.75±.01	.00±.00	.00±.00
	NeuroGPT-D	.51±.06	.73±.01	.00±.00	.00±.00
O	EEGPT	.67±.04	.64±.06	.49±.02	.59±.04
	NeuroLM	.63±.08	.31±.10	.42±.04	.64±.04
	MVPFormer	.56±.03	.56±.03	.37±.04	.43±.04

Table 20: Performance of BFM on the EEGMat.

Type	Model	AUROC	Acc	F1	F2
C	MBrain	.58±.10	.58±.10	.64±.13	.66±.15
	BIOT	.56±.09	.62±.12	.68±.18	.65±.21
	Bendr	.49±.05	.50±.03	.48±.05	.45±.04
	SppEEGNet	.49±.03	.47±.04	.33±.06	.27±.05
G	BrainWave	.88±.07	.81±.06	.83±.08	.81±.11
	REVE	.81±.13	.75±.13	.80±.10	.86±.11
	NeuroGPT-D	.80±.17	.77±.13	.80±.11	.84±.14
	BrainBERT	.73±.11	.66±.15	.63±.26	.62±.29
	NeuroGPT-E	.72±.12	.66±.14	.73±.15	.69±.19
	BrainOmni	.69±.15	.69±.12	.73±.14	.77±.19
	BFM	.69±.15	.62±.05	.73±.04	.81±.08
	CodeBrain	.65±.20	.56±.08	.61±.06	.62±.15
	CBraMod	.54±.14	.58±.05	.58±.18	.58±.24
	LaBraM	.54±.07	.54±.04	.59±.07	.59±.13
	Brant	.51±.11	.46±.08	.47±.17	.50±.30
O	EEGPT	.56±.21	.56±.11	.62±.12	.66±.15
	NeuroLM	.49±.04	.49±.05	.53±.28	.62±.35
	MVPFormer	.48±.05	.51±.04	.53±.09	.54±.13

Table 21: Performance of BFM on the SZ-28.

Type	Model	AUROC	Acc	F1	F2
C	BIOT	.93±.03	.81±.07	.85±.05	.93±.02
	MBrain	.93±.08	.87±.11	.89±.08	.91±.05
	Bendr	.85±.07	.78±.05	.80±.02	.81±.05
	SppEEGNet	.58±.02	.60±.09	.58±.12	.54±.12
G	REVE	.97±.04	.88±.09	.90±.06	.92±.04
	LaBraM	.96±.04	.87±.08	.88±.07	.89±.08
	BrainOmni	.94±.07	.85±.09	.87±.07	.89±.05
	BrainWave	.94±.03	.85±.02	.84±.03	.81±.06
	BFM	.93±.08	.86±.11	.88±.08	.91±.05
	BrainBERT	.92±.07	.85±.07	.85±.06	.85±.10
	Brant	.92±.09	.80±.10	.80±.14	.80±.19
	CBraMod	.90±.13	.84±.12	.86±.09	.87±.09
	NeuroGPT-E	.87±.05	.80±.05	.84±.03	.90±.03
	CodeBrain	.77±.15	.85±.09	.86±.07	.87±.09
	NeuroGPT-D	.70±.12	.67±.13	.77±.07	.89±.04
O	EEGPT	.94±.05	.87±.08	.88±.06	.90±.04
	NeuroLM	.82±.11	.80±.06	.80±.09	.81±.18
	MVPFormer	.73±.09	.83±.06	.84±.04	.86±.06

Table 22: Performance of BFM on the MDD-64.

Type	Model	AUROC	Acc	F1	F2
C	BIOT	.60±.11	.30±.17	.11±.02	.22±.05
	MBrain	.59±.07	.62±.03	.24±.16	.20±.16
	Bendr	.54±.03	.54±.03	.45±.03	.47±.03
	SppEEGNet	.50±.02	.49±.04	.41±.04	.45±.04
G	BrainWave	.72±.03	.66±.03	.62±.03	.67±.06
	BrainOmni	.68±.12	.63±.09	.55±.10	.58±.13
	REVE	.67±.12	.65±.08	.47±.14	.45±.17
	LaBraM	.65±.10	.59±.11	.60±.09	.71±.08
	BrainBERT	.64±.09	.61±.05	.48±.11	.48±.12
	BFM	.62±.10	.59±.06	.48±.10	.51±.13
	NeuroGPT-E	.59±.08	.62±.01	.02±.02	.01±.01
	CBraMod	.54±.05	.69±.14	.13±.08	.16±.10
	Brant	.53±.09	.62±.01	.00±.00	.00±.00
	CodeBrain	.52±.11	.60±.04	.40±.13	.41±.15
	NeuroGPT-D	.51±.08	.62±.01	.00±.00	.00±.00
O	EEGPT	.63±.04	.62±.05	.50±.05	.51±.11
	NeuroLM	.54±.05	.61±.02	.30±.26	.31±.31
	MVPFormer	.51±.02	.55±.02	.39±.06	.38±.09

Table 23: Performance of BFM on the Dep-BDI.

Type	Model	Acc	AUROC	MF1	Kappa
C	BIOT	.37±.07	.56±.09	.33±.07	.073±.088
	MBrain	.60±.03	.55±.04	.30±.05	.054±.059
	Bendr	.48±.03	.54±.03	.36±.03	.047±.041
	SppEEGNet	.46±.03	.51±.02	.33±.01	.007±.024
G	BrainWave	.51±.05	.64±.06	.42±.05	.150±.053
	NeuroGPT-E	.55±.07	.61±.09	.38±.10	.129±.155
	LaBraM	.36±.10	.61±.09	.26±.06	.081±.093
	BrainBERT	.48±.11	.60±.06	.32±.08	.064±.126
	BrainOmni	.53±.05	.60±.06	.36±.05	.105±.062
	BFM	.45±.09	.58±.08	.37±.06	.099±.117
	NeuroGPT-D	.61±.01	.57±.06	.25±.00	.000±.000
	REVE	.54±.10	.54±.11	.35±.08	.079±.200
	CBraMod	.60±.02	.54±.04	.30±.04	.049±.052
	Brant	.61±.01	.53±.03	.25±.00	.000±.000
	CodeBrain	.48±.08	.52±.05	.32±.04	.034±.055
O	EEGPT	.51±.06	.56±.06	.34±.06	.025±.080
	MVPFormer	.50±.04	.54±.03	.34±.01	.028±.039
	NeuroLM	.58±.04	.50±.04	.26±.02	.009±.018

Table 24: Performance of BFM’s on the Dep-STAI.

Type	Model	Acc	AUROC	MF1	Kappa
C	BIOT	.68±.04	.91±.02	.65±.05	.585±.049
	MBrain	.63±.07	.87±.04	.55±.07	.514±.084
	Bendr	.52±.04	.81±.02	.49±.05	.269±.076
	SppEEGNet	.26±.05	.63±.03	.21±.03	.098±.038
G	REVE	.69±.03	.92±.01	.65±.05	.598±.036
	BrainOmni	.68±.03	.91±.02	.64±.04	.581±.030
	CodeBrain	.65±.03	.90±.02	.63±.05	.557±.045
	BrainWave	.62±.05	.89±.01	.60±.02	.522±.064
	CBraMod	.54±.08	.86±.05	.48±.10	.419±.094
	LaBraM	.61±.02	.85±.02	.56±.05	.502±.028
	Brant	.55±.01	.84±.02	.50±.04	.427±.016
	NeuroGPT-E	.44±.11	.80±.02	.41±.11	.319±.101
	BrainBERT	.47±.05	.80±.03	.42±.07	.332±.058
	BFM	.47±.04	.78±.03	.37±.13	.264±.148
	NeuroGPT-D	.31±.07	.61±.07	.17±.08	.060±.102
O	EEGPT	.60±.05	.87±.02	.57±.08	.486±.070
	MVPFormer	.38±.04	.70±.02	.33±.05	.208±.044
	NeuroLM	.27±.11	.67±.05	.15±.10	.074±.074

Table 25: Performance of BFM’s on the ISRUC-G1.

Type	Model	Acc	AUROC	MF1	Kappa
C	BIOT	.78±.03	.95±.01	.74±.02	.699±.043
	MBrain	.78±.04	.95±.01	.74±.03	.709±.059
	Bendr	.69±.06	.91±.01	.65±.04	.591±.062
	SppEEGNet	.43±.11	.76±.03	.39±.07	.283±.075
G	CodeBrain	.73±.07	.94±.02	.71±.05	.646±.087
	NeuroGPT-E	.78±.03	.94±.01	.70±.03	.696±.045
	Brant	.73±.07	.94±.01	.70±.05	.639±.083
	REVE	.75±.04	.94±.02	.71±.04	.664±.048
	CBraMod	.73±.04	.93±.01	.69±.04	.642±.054
	LaBraM	.76±.04	.93±.01	.68±.03	.669±.050
	BrainBERT	.68±.07	.91±.02	.65±.05	.578±.079
	BrainWave	.67±.05	.89±.02	.54±.08	.516±.104
	BFM	.65±.04	.89±.01	.60±.03	.558±.053
	BrainOmni	.63±.03	.87±.00	.60±.01	.506±.025
	NeuroGPT-D	.48±.04	.65±.03	.21±.05	.100±.068
O	EEGPT	.68±.03	.91±.00	.66±.01	.578±.026
	MVPFormer	.49±.08	.81±.02	.45±.06	.339±.068
	NeuroLM	.18±.04	.61±.06	.11±.03	.189±.263

Table 26: Performance of BFM’s on the SleepEDF.

Type	Model	Acc	AUROC	MF1	Kappa
C	MBrain	.27±.02	.54±.02	.19±.03	.027±.023
	SppEEGNet	.26±.01	.51±.01	.30±.02	.115±.012
	BIOT	.26±.01	.51±.01	.24±.01	.009±.012
	Bendr	.26±.01	.51±.01	.26±.01	.013±.013
G	REVE	.34±.02	.66±.05	.30±.02	.115±.012
	NeuroGPT-E	.35±.02	.61±.04	.33±.01	.134±.032
	LaBraM	.29±.02	.56±.04	.22±.03	.047±.021
	CodeBrain	.27±.01	.55±.02	.23±.04	.032±.012
	BrainWave	.27±.02	.54±.02	.20±.03	.031±.022
	BrainOmni	.27±.02	.54±.01	.22±.02	.021±.029
	CBraMod	.27±.01	.53±.01	.22±.03	.024±.020
	Brant	.25±.00	.53±.01	.10±.00	.000±.000
	BFM	.26±.01	.51±.01	.25±.00	.013±.012
	NeuroGPT-D	.26±.01	.51±.03	.15±.03	.007±.019
	BrainBERT	.25±.01	.50±.01	.23±.02	.002±.010
O	EEGPT	.28±.02	.57±.03	.22±.01	.036±.024
	NeuroLM	.27±.01	.52±.02	.20±.02	.030±.018
	MVPFormer	.27±.01	.52±.01	.25±.01	.021±.011

Table 27: Performance of BFM’s on the BCI-2a.

Type	Model	Acc	AUROC	MF1	Kappa
C	Bendr	.29±.01	.55±.01	.29±.01	.054±.010
	MBrain	.26±.01	.51±.01	.16±.05	.009±.011
	SppEEGNet	.26±.01	.51±.01	.23±.01	.008±.013
	BIOT	.25±.00	.50±.01	.20±.02	-.005±.004
G	REVE	.59±.02	.82±.01	.59±.02	.451±.031
	NeuroGPT-E	.54±.01	.78±.01	.54±.01	.392±.019
	BrainOmni	.51±.01	.76±.01	.51±.01	.351±.019
	LaBraM	.31±.02	.59±.02	.25±.03	.078±.032
	CBraMod	.30±.02	.58±.02	.28±.04	.073±.028
	CodeBrain	.30±.02	.58±.02	.28±.03	.068±.020
	BFM	.27±.01	.52±.01	.26±.01	.023±.009
	BrainWave	.25±.01	.51±.01	.18±.01	.001±.014
	Brant	.25±.00	.51±.01	.10±.00	.000±.000
	BrainBERT	.25±.00	.49±.00	.16±.02	-.003±.005
	NeuroGPT-D	.25±.01	.49±.01	.16±.05	-.007±.008
O	MVPFormer	.29±.02	.56±.02	.28±.01	.060±.029
	EEGPT	.27±.02	.52±.01	.24±.04	.023±.022
	NeuroLM	.25±.01	.51±.01	.15±.05	.005±.008

Table 28: Performance of BFM on the EEGMMIDB-I.

Type	Model	Acc	AUROC	MF1	Kappa
C	Bendr	.29±.01	.55±.01	.29±.01	.060±.014
	SppEEGNet	.26±.01	.52±.00	.24±.01	.014±.008
	MBrain	.26±.01	.52±.01	.19±.05	.008±.016
	BIOT	.25±.01	.50±.01	.21±.02	.003±.011
G	REVE	.59±.01	.82±.00	.58±.01	.448±.014
	NeuroGPT-E	.54±.02	.77±.02	.53±.02	.381±.030
	BrainOmni	.49±.01	.74±.01	.49±.01	.325±.014
	CBraMod	.30±.02	.57±.01	.27±.03	.061±.022
	CodeBrain	.30±.02	.57±.01	.27±.02	.067±.026
	LaBraM	.28±.02	.55±.02	.23±.03	.041±.027
	BFM	.27±.01	.52±.01	.26±.01	.024±.008
	NeuroGPT-D	.25±.00	.50±.00	.14±.01	.003±.003
	Brant	.25±.00	.50±.01	.10±.00	.000±.000
	BrainWave	.25±.01	.50±.01	.17±.02	-.004±.011
	BrainBERT	.25±.00	.50±.01	.16±.03	.001±.006
O	MVPFormer	.29±.01	.54±.01	.28±.01	.047±.013
	NeuroLM	.26±.01	.52±.02	.14±.05	.009±.017
	EEGPT	.25±.01	.51±.01	.23±.03	.006±.017

Table 29: Performance of BFM on the EEGMMIDB-R.

Type	Model	Acc	AUROC	MF1	Kappa
C	SppEEGNet	.27±.06	.51±.03	.23±.03	.007±.027
	MBrain	.42±.04	.51±.05	.15±.01	.000±.000
	BIOT	.30±.06	.50±.03	.18±.02	-.014±.030
	Bendr	.26±.02	.50±.02	.24±.02	-.004±.035
G	BrainBERT	.23±.02	.52±.02	.18±.03	.017±.020
	CodeBrain	.27±.12	.52±.03	.16±.06	.004±.027
	NeuroGPT-D	.44±.04	.51±.03	.15±.01	.000±.000
	BrainWave	.27±.07	.51±.03	.23±.06	.010±.077
	CBraMod	.19±.02	.51±.01	.14±.02	-.001±.009
	LaBraM	.29±.04	.51±.02	.21±.03	.016±.014
	REVE	.26±.05	.50±.05	.21±.03	-.013±.035
	BFM	.28±.07	.50±.02	.22±.02	.005±.015
	NeuroGPT-E	.42±.03	.49±.01	.17±.03	-.004±.006
	Brant	.34±.17	.48±.02	.12±.06	.000±.000
	BrainOmni	.22±.03	.45±.03	.19±.01	-.046±.022
O	NeuroLM	.33±.13	.51±.03	.13±.03	-.003±.007
	MVPFormer	.26±.05	.50±.02	.23±.04	.011±.017
	EEGPT	.26±.05	.50±.02	.22±.04	.029±.027

Table 30: Performance of BFM on the DEAP.

Type	Model	Acc	AUROC	MF1	Kappa
C	MBrain	.26±.01	.52±.01	.21±.04	.014±.020
	Bendr	.25±.00	.50±.00	.25±.00	-.001±.005
	SppEEGNet	.25±.00	.50±.00	.24±.00	-.001±.002
	BIOT	.24±.01	.48±.00	.19±.02	-.018±.003
G	NeuroGPT-E	.30±.01	.57±.01	.26±.02	.003±.005
	REVE	.28±.02	.54±.01	.27±.01	.039±.019
	CodeBrain	.26±.01	.53±.02	.20±.03	.020±.014
	BrainOmni	.28±.02	.53±.01	.26±.01	.035±.019
	BrainWave	.27±.02	.53±.02	.22±.03	.023±.026
	BFM	.27±.01	.52±.01	.24±.01	.012±.004
	CBraMod	.26±.01	.51±.01	.21±.03	.010±.010
	BrainBERT	.25±.02	.51±.03	.17±.05	.003±.012
	LaBraM	.25±.01	.51±.01	.20±.02	.008±.014
	NeuroGPT-D	.27±.00	.50±.00	.11±.00	.000±.000
	Brant	.26±.01	.50±.01	.10±.00	.000±.000
O	EEGPT	.25±.01	.51±.01	.22±.02	.006±.016
	MVPFormer	.25±.01	.51±.00	.23±.02	.005±.006
	NeuroLM	.25±.00	.50±.00	.13±.04	.001±.002

Table 31: Performance of BFM on the SEED-IV.

Type	Model	Acc	AUROC	MF1	Kappa
C	MBrain	.03±.00	.50±.01	.00±.00	.000±.001
	Bendr	.03±.00	.50±.01	.02±.00	.000±.002
	SppEEGNet	.02±.00	.50±.00	.01±.00	-.000±.001
	BIOT	.03±.01	.49±.00	.01±.00	.000±.001
G	LaBraM	.04±.01	.51±.01	.01±.00	.005±.004
	CBraMod	.04±.01	.51±.01	.01±.00	.003±.003
	CodeBrain	.05±.00	.51±.01	.00±.00	.001±.001
	BrainOmni	.02±.00	.50±.01	.01±.00	.001±.001
	BFM	.03±.00	.50±.00	.01±.00	.001±.001
	NeuroGPT-D	.05±.00	.50±.00	.00±.00	.001±.004
	BrainBERT	.03±.00	.50±.01	.01±.01	.001±.003
	REVE	.03±.00	.50±.00	.02±.00	.000±.001
	Brant	.04±.01	.50±.00	.00±.00	.000±.000
	NeuroGPT-E	.05±.00	.50±.01	.00±.00	-.000±.004
	BrainWave	.03±.01	.50±.01	.00±.00	.002±.003
O	EEGPT	.03±.01	.50±.00	.01±.00	.001±.003
	NeuroLM	.04±.01	.50±.00	.00±.00	-.001±.001
	MVPFormer	.02±.00	.50±.01	.02±.00	.001±.002

Table 32: Performance of BFM on the Chisco-I.

Type	Model	Acc	AUROC	MF1	Kappa
C	BIOT	.04±.00	.50±.01	.01±.00	-.000±.003
	MBrain	.04±.00	.50±.00	.00±.00	.000±.000
	SppEEGNet	.02±.00	.50±.00	.01±.00	.000±.003
	Bendr	.03±.00	.50±.00	.02±.00	.001±.003
G	BrainWave	.03±.01	.51±.02	.00±.00	.001±.003
	BrainOmni	.03±.01	.51±.01	.01±.00	.002±.005
	LaBraM	.03±.01	.51±.01	.01±.00	.002±.004
	NeuroGPT-D	.05±.00	.51±.01	.00±.00	.002±.003
	CBraMod	.03±.01	.51±.01	.01±.00	-.000±.005
	NeuroGPT-E	.05±.00	.50±.01	.00±.00	.001±.003
	BFM	.03±.01	.50±.00	.01±.00	.001±.002
	REVE	.03±.00	.50±.01	.02±.00	.000±.003
	Brant	.04±.01	.50±.00	.00±.00	.000±.000
	BrainBERT	.03±.01	.50±.00	.00±.00	.001±.001
	CodeBrain	.03±.01	.50±.01	.01±.00	-.003±.001
O	EEGPT	.03±.01	.50±.01	.01±.01	.002±.002
	NeuroLM	.03±.02	.50±.00	.00±.00	-.000±.000
	MVPFormer	.02±.00	.50±.00	.02±.00	.001±.002

Table 33: Performance of BFM on the Chisco-R.

Frozen Encoder Fine-tuning

In frozen-FT, the pretrained encoder is fixed and only the downstream classification head is optimized. For each model-dataset pair, we use the classifier-head learning rate selected in the corresponding full-FT setting, avoiding an arbitrary fixed head LR while keeping the two settings comparable. We additionally conduct a head-LR sensitivity analysis to examine whether the shared setting unfairly disadvantages frozen-FT. We evaluate representative model-dataset pairs with large full-FT gains, covering Bendr, EEGPT, SppEEGNet, and NeuroGPT on MDD-64, MAYO, ADHD-Adult, and ISRUC-G1. For most pairs, varying the head LR produces only minor changes in frozen-FT performance and therefore

Type	Model	AUROC	Acc	F1	F2
C	SppEEGNet	.51±.02	.50±.01	.44±.07	.42±.09
	MBrain	.50±.01	.50±.01	.63±.07	.76±.15
	BIOT	.50±.02	.50±.01	.34±.32	.38±.39
G	LaBraM	.53±.01	.50±.00	.53±.30	.66±.37
	CodeBrain	.52±.01	.52±.01	.29±.23	.28±.32
	BrainBERT	.52±.01	.50±.00	.00±.00	.00±.00
	CBraMod	.50±.01	.50±.00	.24±.28	.25±.35
	BFM	.50±.01	.50±.00	.51±.05	.52±.09
	BrainWave	.49±.01	.50±.00	.53±.30	.67±.37
O	MVPFormer	.51±.00	.49±.00	.53±.05	.57±.08

Table 34: Performance of BFM on the Cogitate.

cannot explain the advantage of full-FT. EEGPT on MAYO is the only clearly LR-sensitive case. Nevertheless, the head LR used in the main experiment (5×10^{-4}) yields a validation AUROC of 73.20, close to the best searched result of 75.13 at 10^{-3} and still far below the full-FT result of 91.76. These results indicate that the shared head-LR setting does not materially disadvantage frozen-FT or alter its performance ordering relative to full-FT. Figure 5(a) reports the difference in mean AUROC between full-FT and frozen-FT for each model-dataset pair, with significance indicated by asterisks.

Additional Prototype-based Transfer Analysis

Figure 5(c) provides the detailed results underlying the pair-level heterogeneity summarized in the main text, showing the AUROC distributions of seven models on six datasets under the 1-, 3-, 8-, and 15-shot settings, with each box summarizing all fold-level results from five repeated support-set samplings per fold. To assess whether the global shot-performance trend is consistent across model-dataset combinations, we conducted pair-level analyses. The results show clear pair-level heterogeneity. Among the 42 model-dataset pairs with all four shot settings available, 22 exhibit a significant positive 1/3/8/15-shot trend after FDR correction; for the 15-shot versus 1-shot comparison, 21 of 42 pairs show a significant improvement for 15-shot. Thus, although the aggregate results indicate a positive shot-performance relationship, statistically reliable gains are limited to a subset of model-dataset pairs. This heterogeneity is also visible in the pair-level patterns. Several models on SleepEDF show significant positive trends together with relatively large 15-shot gains, while BrainWave shows particularly large improvements on ADHD-Adult, SZ-28, and ADFD. In contrast, other pairs show only modest or non-significant gains, including ADFD/BrainOmni, ADFD/MBrain, SZ-28/REVE, ADHD-Adult/CBraMod, and MDD-64/CBraMod.

Cross-modal representation geometry

This section provides the full protocol for the cross-modal representation analysis in **EEG and iEEG modalities**. The analysis examines whether pretrained BFM encode the two modalities in shared principal subspaces, whether this sharing is directionally asymmetric, and whether modality-

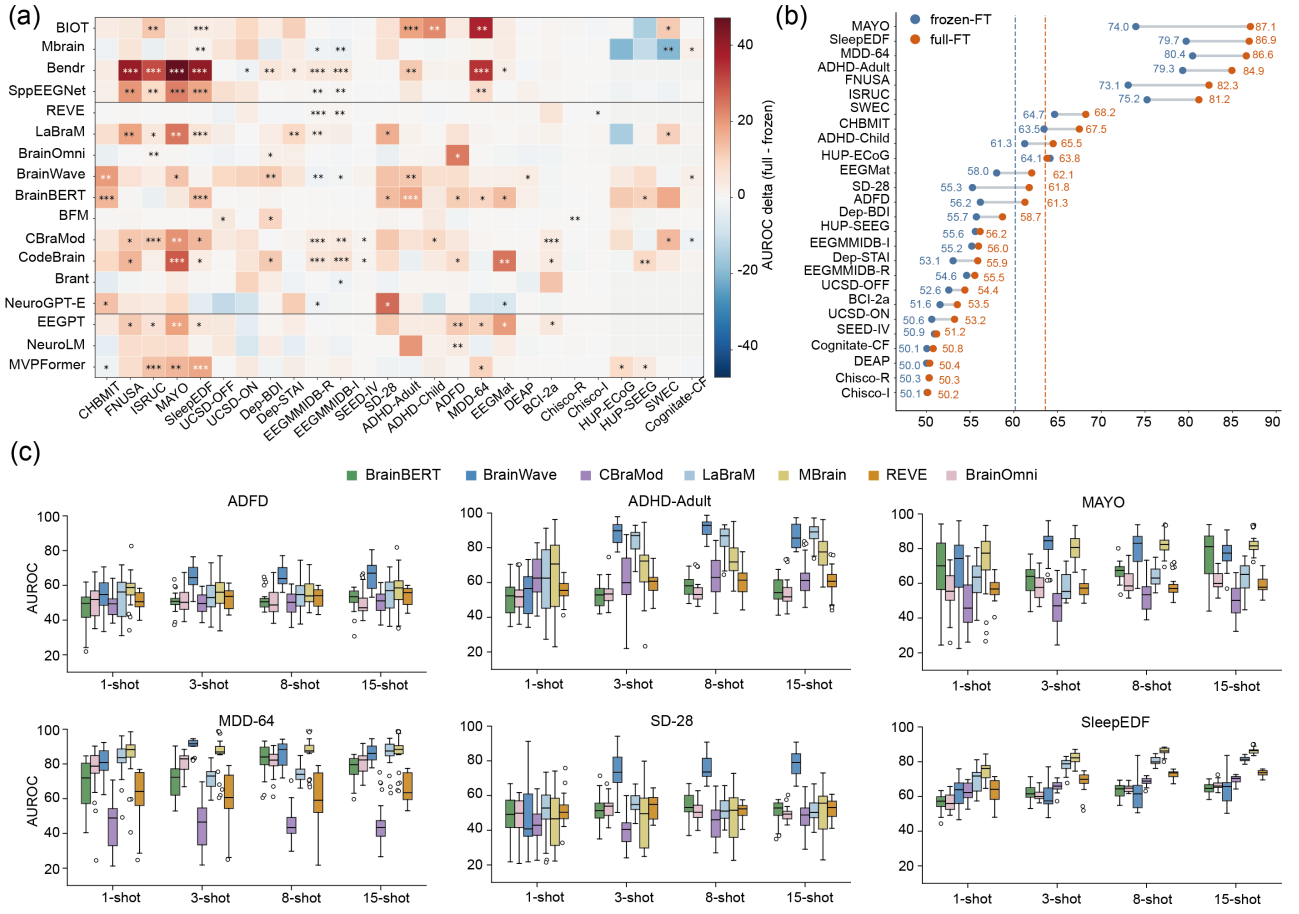


Figure 5: Effect of frozen-FT and full-FT. (a) Validation AUROC gain of full-FT over frozen-FT for each model-dataset pair, computed as average $\text{AUROC}_{\text{full}} - \text{average AUROC}_{\text{frozen}}$ (* $p < 0.05$, ** $p < 0.01$, and *** $p < 0.001$). (b) Dataset-level average validation AUROC across all models under frozen-FT and full-FT. (c) Prototype-based few-shot evaluation.

specific residual structure is associated with frozen-transfer performance. The iEEG pool was fixed across replicates and comprised HUP-ECOG, HUP-SEEG, MAYO, FNUSA, SWEC, and Cogitate-CF. We used three EEG-cohort draws: ADHD-Adult, ADHD-Child, BCI-2a, CHBMIT, EEGMMIDB-I, and ISRUC-G1 (seed 2026); BCI-2a, CHBMIT, Chisco-R, DEAP, EEGMMIDB-R, and MDD-64 (seed 2030); and ADFD, Chisco-I, DEAP, EEGMMIDB-I, ISRUC, and SleepEDF (seed 2089).

For each model-dataset pair, we load the original pre-trained checkpoint, remove the prediction head, and extract sample-level representations from matched 2-s epochs. Each cohort contributes 2,000 label-balanced samples. Representations are centered separately within each cohort before pooling, reducing cohort-level mean shifts associated with task and acquisition differences. The EEG and iEEG pools therefore contain 12,000 samples each. For the mixed pool, we sample 1,000 epochs from each of the 12 cohorts, yielding the same total sample size.

Let $\mathbf{X}_M \in \mathbb{R}^{n_M \times d}$ denote the pooled centered representations for $M \in \{\text{EEG}, \text{iEEG}, \text{mix}\}$, where d is the hidden

dimension. We compute the covariance matrix

$$\Sigma_M = \frac{1}{n_M - 1} \mathbf{X}_M^\top \mathbf{X}_M \quad (1)$$

and its eigendecomposition

$$\Sigma_M = \mathbf{V}_M \Lambda_M \mathbf{V}_M^\top, \quad (2)$$

with eigenvalues $\lambda_{M,1} \geq \dots \geq \lambda_{M,D}$. For $\alpha = 0.99$, we define the effective PCA dimensionality as

$$D_\alpha^M = \min \left\{ k : \frac{\sum_{i=1}^k \lambda_{M,i}}{\sum_{i=1}^D \lambda_{M,i}} \geq \alpha \right\}. \quad (3)$$

Let $\mathbf{U}_M \in \mathbb{R}^{d \times D_{99}^M}$ denote the leading eigenvectors spanning the corresponding 99%-variance subspace.

To obtain an out-of-sample estimate of subspace coverage, we split each pooled modality matrix into complementary interleaved halves, $\mathbf{X}_M^{(a)}$ and $\mathbf{X}_M^{(b)}$. PCA subspaces are fitted independently on each half. For source modality S and target modality T , directional coverage is evaluated on the opposite

Table 35: Cross-modal geometry of pretrained representations after dataset-wise centering. $C_{E \rightarrow I}$ denotes the fraction of iEEG variance explained by the EEG 99% PCA subspace, and $C_{I \rightarrow E}$ is defined symmetrically.

Model	D_{99}^{EEG}	D_{99}^{iEEG}	D_{99}^{mix}	$C_{E \rightarrow I}$	$C_{I \rightarrow E}$	$\bar{C}$
BrainWave	22.0 ± 5.2	11.0 ± 0.0	22.0 ± 2.6	0.940 ± 0.033	0.866 ± 0.007	0.903 ± 0.017
BrainBERT	184.0 ± 5.6	173.0 ± 0.0	186.0 ± 2.6	0.986 ± 0.001	0.981 ± 0.006	0.984 ± 0.003
BIOT	49.7 ± 5.5	47.0 ± 0.0	58.7 ± 3.1	0.970 ± 0.005	0.969 ± 0.003	0.970 ± 0.002
LaBraM	128.0 ± 4.0	117.0 ± 0.0	130.3 ± 1.5	0.976 ± 0.004	0.969 ± 0.001	0.973 ± 0.002
CBraMod	64.7 ± 11.2	83.0 ± 0.0	80.0 ± 6.0	0.969 ± 0.006	0.983 ± 0.005	0.976 ± 0.002
SppEEGNet	37.7 ± 0.6	37.0 ± 0.0	38.0 ± 0.0	0.987 ± 0.002	0.985 ± 0.001	0.986 ± 0.001
Mbrain	351.3 ± 38.7	495.0 ± 0.0	508.3 ± 6.7	0.805 ± 0.041	0.885 ± 0.017	0.845 ± 0.028
CodeBrain	93.7 ± 3.8	93.0 ± 0.0	95.3 ± 1.5	0.983 ± 0.004	0.985 ± 0.001	0.984 ± 0.001
MVPFormer	452.3 ± 35.5	316.0 ± 0.0	414.0 ± 21.5	0.987 ± 0.006	0.952 ± 0.011	0.970 ± 0.008
BFM	129.0 ± 30.5	94.0 ± 0.0	126.0 ± 14.0	0.979 ± 0.005	0.928 ± 0.042	0.953 ± 0.022

target half and averaged over the two split directions:

$$C_{S \rightarrow T} = \frac{1}{2} \left(\frac{\|\mathbf{X}_T^{(b)} \mathbf{U}_S^{(a)}\|_F^2}{\|\mathbf{X}_T^{(b)}\|_F^2} + \frac{\|\mathbf{X}_T^{(a)} \mathbf{U}_S^{(b)}\|_F^2}{\|\mathbf{X}_T^{(a)}\|_F^2} \right). \quad (4)$$

Here, $\mathbf{U}_S^{(h)}$ is the orthonormal 99%-variance PCA basis fitted on source split h . Thus, $C_{S \rightarrow T}$ measures the fraction of target variance retained after projection onto the source subspace. We compute both $C_{\text{EEG} \rightarrow \text{iEEG}}$ and $C_{\text{iEEG} \rightarrow \text{EEG}}$ to characterize directional asymmetry.

Within-modality coverage is computed identically using complementary halves of the same modality:

$$C_{M \rightarrow M} = \frac{1}{2} \left[\frac{\|\mathbf{X}_M^{(b)} \mathbf{U}_M^{(a)}\|_F^2}{\|\mathbf{X}_M^{(b)}\|_F^2} + \frac{\|\mathbf{X}_M^{(a)} \mathbf{U}_M^{(b)}\|_F^2}{\|\mathbf{X}_M^{(a)}\|_F^2} \right]. \quad (5)$$

We define the modality-specific residual gap as the mean loss in coverage when replacing a within-modality subspace with the opposite-modality subspace:

$$G = \frac{1}{2} \left[(C_{\text{EEG} \rightarrow \text{EEG}} - C_{\text{iEEG} \rightarrow \text{EEG}}) + (C_{\text{iEEG} \rightarrow \text{iEEG}} - C_{\text{EEG} \rightarrow \text{iEEG}}) \right]. \quad (6)$$

Larger G indicates that within-modality subspaces retain substantially more variance than cross-modal subspaces, corresponding to stronger modality-specific residual structure. We report the mean and standard deviation across the three EEG-cohort draws and relate G to macro-averaged frozen-transfer AUROC across the cohorts included in each draw.

Additional Spatial Analysis Results

This subsection provides detailed results supporting the analysis discussed in **Spatial Sensitivity**. We report the full quantitative impact of channel-permutation perturbations across datasets and models. We permute channels in the training and validation splits to probe whether BFMs encode dataset-specific spatial structure. For models using channel identifiers, signal channels and their corresponding identifiers are jointly permuted to preserve signal-identifier correspondence. All other settings are kept identical, so performance

changes reflect sensitivity to the disrupted spatial organization, including channel-wise heterogeneity and, for order-dependent models, induced topology. Tables 36 to 40 report all metrics before and after permutation. We focus on AUROC in the main analysis, and these results support that some BFMs internalize dataset-specific spatial structure during training.

	AUROC		Acc		F1		F2	
Model	shuffle	origin	shuffle	origin	shuffle	origin	shuffle	origin
BrainBERT	.54±.04	.55±.03	.55±.03	.56±.03	.58±.04	.56±.07	.58±.05	.53±.09
BrainWave	.68±.04	.78±.04	.66±.05	.71±.04	.68±.12	.74±.06	.68±.19	.74±.12
BrainOmni	.71±.09	.71±.09	.62±.07	.62±.07	.63±.09	.63±.09	.61±.15	.61±.14
CBraMod	.60±.06	.64±.05	.57±.03	.64±.06	.72±.03	.73±.03	.85±.03	.81±.05
EEGPT	.63±.11	.67±.13	.60±.10	.64±.10	.57±.28	.68±.10	.59±.32	.68±.12
MBrain	.69±.11	.69±.11	.63±.09	.63±.08	.68±.07	.66±.06	.70±.05	.66±.04
NeuroGPT-E	.55±.02	.55±.02	.56±.03	.56±.03	.72±.02	.72±.02	.86±.01	.86±.01
REVE	.71±.09	.79±.07	.55±.11	.71±.04	.36±.31	.74±.06	.32±.30	.74±.08

Table 36: Performance under channel-permutation setting. We report AUROC, Accuracy, F1 and F2 for models trained on original data and shuffled data on ADHD-Child.

	AUROC		Acc		F1		F2	
Model	shuffle	origin	shuffle	origin	shuffle	origin	shuffle	origin
BrainBERT	.62±.05	.59±.05	.64±.01	.58±.05	.66±.02	.60±.11	.64±.04	.64±.15
BrainWave	.62±.12	.74±.05	.57±.09	.68±.05	.49±.26	.68±.10	.47±.31	.73±.10
BrainOmni	.80±.07	.80±.07	.71±.04	.71±.04	.73±.05	.73±.05	.72±.09	.72±.09
CBraMod	.51±.02	.55±.06	.52±.07	.59±.04	.32±.26	.41±.20	.28±.25	.37±.20
EEGPT	.79±.04	.81±.05	.72±.04	.72±.05	.72±.05	.73±.05	.68±.07	.70±.07
MBrain	.52±.13	.53±.12	.51±.09	.50±.10	.54±.12	.53±.12	.54±.15	.52±.14
NeuroGPT-E	.49±.05	.51±.03	.55±.01	.55±.01	.71±.01	.71±.01	.85±.01	.85±.01
REVE	.79±.08	.84±.07	.68±.07	.77±.08	.74±.05	.79±.07	.79±.07	.79±.09

Table 37: Channel-permutation performance on ADFD.

Additional Frequency Analysis Results

This subsection provides detailed results supporting the frequency-band analysis described in **Frequency-Domain Associations**. Our goal is not to isolate spectral information acquired during pretraining, but to characterize the frequency structure retained after downstream adaptation. We report the full quantitative outcomes of the band-wise PSD prediction experiments. Tables 41 to 43 report the band-wise prediction

	AUROC		Acc		F1		F2	
Model	shuffle	origin	shuffle	origin	shuffle	origin	shuffle	origin
BrainBERT	.79±.04	.78±.06	.80±.04	.75±.10	.59±.08	.58±.08	.59±.11	.61±.09
BrainWave	.59±.14	.90±.03	.63±.09	.80±.12	.30±.11	.71±.09	.32±.15	.78±.04
BrainOmni	.74±.13	.74±.13	.67±.19	.67±.19	.51±.12	.51±.12	.56±.12	.56±.12
CBraMod	.67±.12	.69±.04	.67±.11	.70±.08	.42±.16	.46±.08	.48±.23	.49±.14
EEGPT	.69±.13	.71±.09	.73±.10	.67±.11	.47±.16	.46±.06	.48±.19	.51±.15
MBrain	.70±.08	.75±.08	.77±.03	.70±.11	.22±.22	.54±.10	.17±.18	.59±.09
NeuroGPT-E	.65±.07	.78±.12	.70±.06	.75±.07	.26±.10	.51±.22	.24±.14	.63±.16
REVE	.78±.12	.84±.10	.59±.25	.78±.12	.53±.18	.63±.14	.65±.12	.66±.13

Table 38: Channel-permutation performance on CHBMIT.

	AUROC		Acc		F1		F2	
Model	shuffle	origin	shuffle	origin	shuffle	origin	shuffle	origin
BrainBERT	.50±.09	.51±.05	.49±.05	.51±.06	.57±.04	.51±.11	.64±.06	.54±.17
BrainWave	.50±.11	.69±.10	.49±.12	.53±.08	.40±.28	.60±.06	.43±.31	.69±.14
BrainOmni	.51±.19	.64±.09	.46±.15	.57±.06	.44±.21	.54±.16	.46±.26	.57±.26
CBraMod	.59±.08	.60±.12	.51±.07	.49±.08	.21±.29	.18±.08	.22±.35	.13±.07
EEGPT	.63±.17	.51±.21	.55±.14	.48±.17	.49±.16	.49±.15	.46±.16	.49±.14
MBrain	.43±.16	.49±.15	.46±.06	.46±.14	.49±.21	.37±.17	.56±.28	.34±.18
NeuroGPT-E	.55±.25	.56±.10	.53±.10	.53±.07	.51±.25	.58±.10	.57±.34	.64±.13
REVE	.58±.12	.63±.19	.58±.06	.56±.13	.53±.17	.45±.29	.53±.25	.46±.33

Table 39: Channel-permutation setting performance on UCSD-ON.

	AUROC		Acc		F1		F2	
Model	shuffle	origin	shuffle	origin	shuffle	origin	shuffle	origin
BrainBERT	.52±.09	.51±.06	.53±.06	.54±.06	.60±.10	.49±.17	.68±.15	.49±.20
BrainWave	.49±.19	.58±.06	.56±.10	.53±.10	.43±.29	.62±.12	.44±.34	.71±.19
BrainOmni	.62±.07	.64±.09	.56±.04	.57±.06	.57±.10	.54±.16	.61±.21	.57±.26
CBraMod	.60±.05	.63±.05	.56±.04	.58±.04	.32±.17	.46±.09	.27±.19	.40±.12
EEGPT	.62±.15	.61±.13	.60±.12	.55±.08	.55±.17	.56±.10	.53±.20	.58±.17
MBrain	.48±.09	.50±.04	.44±.06	.44±.04	.35±.22	.32±.23	.38±.31	.34±.29
NeuroGPT-E	.56±.07	.56±.06	.52±.05	.54±.05	.57±.09	.57±.12	.63±.17	.62±.22
REVE	.63±.11	.62±.19	.59±.13	.58±.17	.61±.13	.58±.19	.63±.16	.60±.24

Table 40: Channel-permutation setting performance on UCSD-OFF.

performance for all evaluated models across the six canonical frequency bands. For each band b , we list the Pearson correlation between the ground-truth PSD and the predicted PSD, as well as the normalized relative strength r_b^n used in the main analysis.

Codebook Analysis Results

This subsection provides the complete results supporting the **Codebook Representation** in the main text. We first report codebook token usage and class geometry across all evaluated settings. We then present a detailed analysis of LaBraM on ADFD, CHBMIT, SZ-28, and SleepEDF, comparing standard fine-tuning without the codebook with codebook-enabled fine-tuning under identical protocols in terms of AUROC, accuracy, F1, and F2.

	ADFD						Dep-BDI					
Model	r_d^n	r_t^n	r_a^n	r_b^n	r_{gl}^n	r_{gh}^n	r_d^n	r_t^n	r_a^n	r_b^n	r_{gl}^n	r_{gh}^n
REVE	.85	.00	.71	.11	.59	1.0	.99	.00	1.0	.92	.78	.65
EEGPT	1.0	.32	.36	.20	.92	.00	.80	.67	1.0	.67	.13	.00
BrainOmni	.95	.00	.82	.02	.28	1.0	.83	.39	1.0	.81	.35	.00
BrainWave	.63	.35	.77	.00	.14	1.0	1.0	.47	.75	.23	.00	.28
LaBraM	.08	.00	1.0	.26	.28	.10	.77	.36	1.0	.55	.15	.00
BrainBERT	.69	.08	.16	.00	.68	1.0	.70	.44	1.0	.00	.27	.87
BFM	.89	.00	.33	.09	.27	1.0	.89	1.0	.99	.76	1.0	.71
BIOT	.87	.00	.62	.31	.37	1.0	.80	.38	1.0	.75	.17	.00
CBraMod	.84	.00	.67	.55	.64	1.0	.91	.00	1.0	.86	.47	.05
MBrain	.87	.00	.42	.15	.40	1.0	.98	.00	1.0	.87	.87	.61
SppEEGNet	.26	.10	.00	.02	.27	1.0	1.0	.55	.94	.27	.00	.08
NeuroLM	.66	.89	.51	.08	.00	1.0	1.0	.36	.35	.00	.20	.05
NeuroGPT-E	.85	.13	.00	.01	.06	1.0	.52	.39	1.0	.00	.06	.10
Bendr	.74	.28	.00	.10	.07	1.0	1.0	.28	.79	.44	.00	.03
Brant	.53	.33	.17	.08	.00	1.0	.00	.68	.33	.43	.93	1.0
NeuroGPT-D	.95	.23	.25	.10	.00	1.0	1.0	.30	.86	.62	.11	.00

Table 41: Band-wise PSD predictability across BFM on ADFD and Dep-BDI, reporting normalized band predictability r_b^n for delta (d), theta (t), alpha (a), beta (b), low gamma (gl), and high gamma (gh).

	SZ-28						MDD-64					
Model	r_d^n	r_t^n	r_a^n	r_b^n	r_{gl}^n	r_{gh}^n	r_d^n	r_t^n	r_a^n	r_b^n	r_{gl}^n	r_{gh}^n
BrainWave	.95	.64	1.0	.75	.00	.22	.89	1.0	.89	.61	.00	.02
REVE	1.0	.00	.97	.66	.67	.22	.96	1.0	.99	1.0	.93	.00
NeuroGPT-E	1.0	.87	.06	.00	.41	.90	.25	1.0	.00	.62	.60	.08
BrainBERT	.82	.15	1.0	.00	.13	.29	.75	1.0	.73	.70	.80	.00
BrainOmni	.86	.46	1.0	.71	.27	.00	.66	.84	1.0	.99	.74	.00
BFM	.92	.00	1.0	.84	.90	.84	.86	1.0	.97	.97	.97	.00
MBrain	1.0	.00	1.0	.94	.83	.69	.94	.96	.96	1.0	.92	.00
EEGPT	.91	.66	1.0	.97	.38	.00	.89	.97	1.0	.89	.74	.00
BIOT	.68	.00	1.0	.58	.65	.19	.93	.97	1.0	.99	.86	.00
NeuroGPT-D	1.0	.00	.76	.67	.36	.14	.92	1.0	.98	.71	.67	.00
CBraMod	1.0	.25	.90	.89	.58	.00	.95	.96	.98	1.0	.90	.00
LaBraM	.87	.44	1.0	.70	.34	.00	.59	.86	.59	1.0	.76	.00
Brant	.64	.00	1.0	.71	.61	.54	.84	.87	.89	1.0	.93	.00
Bendr	1.0	.34	.53	.51	.01	.00	.73	1.0	.48	.49	.46	.00
SppEEGNet	1.0	.16	.68	.29	.00	.02	1.0	.84	.78	.63	.51	.00
NeuroLM	1.0	.31	.81	.31	.00	.24	.59	.86	.59	1.0	.76	.00

Table 42: Band-wise PSD predictability on SZ-28 and MDD-64.

Model	ADHD-Adult						ADHD-Child					
	r_d^n	r_t^n	r_a^n	r_b^n	r_{gl}^n	r_{gh}^n	r_d^n	r_t^n	r_a^n	r_b^n	r_{gl}^n	r_{gh}^n
EEGPT	.66	.00	.59	1.0	.18	.08	.95	.82	1.0	1.0	.56	.00
BrainOmni	.65	.00	.67	1.0	.32	.14	1.0	.86	.99	.84	.39	.00
BrainWave	.32	.00	.68	1.0	.80	.82	.83	.59	.94	1.0	.34	.00
BIOT	.40	.00	.40	1.0	.58	.17	.98	.89	1.0	.78	.32	.00
REVE	.71	.00	.25	1.0	.53	.30	.96	.82	.98	1.0	.62	.00
MBrain	.66	.00	.55	1.0	.53	.16	1.0	.87	.98	1.0	.60	.00
Brant	.71	.00	.33	1.0	.32	.08	.89	.58	.68	1.0	.59	.00
CBraMod	.65	.19	.61	1.0	.52	.00	.99	.86	.94	1.0	.79	.00
LaBraM	.76	.39	.71	1.0	.08	.00	1.0	.86	.96	.92	.41	.00
NeuroGPT-E	.39	.74	.00	1.0	.40	.15	.84	1.0	.75	.37	.00	.44
NeuroGPT-D	.95	.00	.71	1.0	.16	.03	1.0	.70	.83	.77	.34	.00
BFM	.61	.00	.45	1.0	.70	.62	.96	.83	.94	1.0	.67	.00
NeuroLM	.41	.00	.55	1.0	.29	.19	1.0	.82	.97	.88	.27	.00
BrainBERT	1.0	.26	.91	.84	.13	.00	1.0	.64	.82	.66	.60	.00
SppEEGNet	1.0	.06	.72	.61	.06	.00	1.0	.80	.70	.64	.31	.00
Bendr	1.0	.16	.73	.78	.03	.00	1.0	.62	.83	.57	.20	.00

Table 43: Band-wise PSD predictability across BFM on ADHD.

Model	ADFD				CHBMIT				SZ-28			
	Cov.	Ent	Inter	DR	Cov.	Ent	Inter	DR	Cov.	Ent	Inter	DR
BFM	.430	.868	.998	.002	.635	.838	.999	.001	.547	.853	.999	.001
BrainOmni ₁	.997	.947	.996	.007	.988	.948	.999	.001	.964	.941	.999	.002
BrainOmni ₂	.998	.891	.928	.077	.998	.950	.966	.037	.998	.982	.994	.007
BrainOmni ₃	.998	.929	.927	.076	.998	.970	.959	.043	.998	.973	.954	.049
BrainOmni ₄	.998	.946	.947	.055	.998	.988	.930	.072	.998	.972	.909	.094
LaBraM _{no}	.353	.755	.995	.006	.369	.721	.973	.038	.171	.607	.986	.022
LaBraM	.140	.694	.267	1.155	.093	.518	-.954	6.755	.043	.586	.800	.294

Table 44: Codebook utilization and class geometry under different settings. We report codebook coverage (Cov), normalized token entropy (Ent), mean inter-class similarity (Inter), and the inter-/intra-class distance ratio (DR) for BFM, LaBraM with a frozen or updated codebook, and the four RVQ levels of BrainOmni.

Dataset	AUROC		Acc		F1		F2	
	CB	origin	CB	origin	CB	origin	CB	origin
ADFD	.65±.06	.77±.10	.62±.05	.72±.08	.68±.03	.75±.06	.70±.03	.75±.06
CHBMIT	.77±.06	.78±.07	.75±.04	.75±.06	.60±.10	.54±.24	.55±.07	.53±.16
SZ-28	.47±.12	.54±.07	.50±.09	.54±.04	.56±.12	.59±.07	.60±.15	.59±.13

Table 45: LaBraM codebook analysis on ADFD, CHBMIT, and SZ-28. We compare standard fine-tuning without codebook (Origin) and codebook-enabled fine-tuning (CB) under identical protocols, reporting AUROC, accuracy, F1, and F2.

AUROC		Acc		MF1		Kappa	
CB	origin	CB	origin	CB	origin	CB	origin
.86±.02	.93±.01	.66±.04	.76±.04	.61±.03	.68±.03	.54±.05	.67±.05

Table 46: LaBraM codebook analysis on SleepEDF, reporting Accuracy, AUROC (OvR), macro-F1 (MF1), and Cohen’s κ .

SSL Paradigms

EEG and iEEG are naturally modeled as multivariate time series. We denote a recording as $X \in \mathbb{R}^{C \times N}$, where C is the number of channels and $N = f_s \times t$ is the number of samples over duration t at sampling rate f_s . For electrode-wise representations, let $\epsilon \subseteq \{1, \dots, C\}$ be a subset of electrodes with $E_\epsilon = |\epsilon|$. The corresponding multivariate time series is $x^\epsilon \in \mathbb{R}^{E_\epsilon \times N}$. We categorize pretraining pipelines into patch-level and sequence-level formulations. In patch-level modeling, the raw signal X is partitioned along the temporal dimension into k non-overlapping windows of length $w = \frac{N}{k}$. Objectives are defined at the patch granularity, and patches are typically treated as exchangeable tokens. Sequence-level modeling preserves the temporal structure and treats the input as an ordered sequence.

SSL derives training signals from the data, enabling pre-training without manual annotations. Most SSL methods can be viewed as optimizing an objective constructed from transformations and prediction modules applied to the input signal. We form two sets from each input, a source set $\mathcal{X}_{src}$ and a target set $\mathcal{X}_{tgt}$. An encoder $\mathcal{F}(\cdot)$ maps the input to a d -dimensional latent representation, followed by an optional module $\mathcal{G}(\cdot)$ that serves either as a decoder for reconstruction or as a prediction or alignment head. Under this formulation, SSL methods can be expressed with a unified objective:

$$\mathcal{L}_{\text{sum}} = \mathcal{L}(\mathcal{G}(\mathcal{Q}(\mathcal{F}(\mathcal{T}_{d,s}(\mathcal{X}_{src}), \mathcal{M}))), \mathcal{F}(\mathcal{T}_{d,s}(\mathcal{X}_{tgt})), \mathcal{N}) \quad (7)$$

Here, $\mathcal{T}_{d,s} = \Phi_d \circ a_s$ denotes a deterministic transform composed of domain mapping Φ_d and augmentation a_s . The tokenizer $\mathcal{Q}(\cdot)$ discretizes continuous latent representations. And $\mathcal{M}$ denotes a masking operator, while $\mathcal{N}$ is the negative set when contrastive objectives are employed. Based on this formulation, we categorize BFM pretraining strategies into three SSL paradigms.

Contrastive-based Method

Contrastive-based SSL defines learning objectives by comparing representations of paired inputs and contrasting them against a set of negatives. This paradigm is expressed as:

$$\mathcal{L}_C = \mathcal{L}(\mathcal{G}(z_1), z_2, \mathcal{N}) \quad (8)$$

where $z_1 = \mathcal{F}_1(\mathcal{T}_{d,s}(\mathcal{X}_{src}, \mathcal{M}))$ and $z_2 = \mathcal{F}_2(\mathcal{T}_{d,s}(\mathcal{X}_{tgt}))$, $\mathcal{F}_1(\cdot)$ produces a query representation from the causal context while $\mathcal{F}_2(\cdot)$ generates the target embedding from a future segment. The projection head $\mathcal{G}(\cdot)$ maps representations into a latent space. The loss contrasts the positive pair (z_1, z_2) with negatives $\mathcal{N}$, promoting alignment of matched pairs and separation from mismatches.

Augmentation contrast Augmentation-based contrastive learning constructs positive pairs by applying transformation operators a_s to the same input signal, with representation learning driven by enforcing invariance across augmented views. Early BFM studies primarily focused on expanding the effective view space induced by a_s . SeqCLR (Mohsenvand, Izadi, and Maes 2020) adapts SimCLR (Chen et al. 2020) to multichannel EEG by recombining channels to expand the augmentation-induced view space and enforce consistency across views. SppEEGNet (Li and Metsis 2022) further

applies a suite of EEG-specific augmentations and forms positive pairs within sliding windows on a 2D signal representation, which helps reduce cross-dataset sampling-rate mismatch. BIOT (Yang, Westover, and Sun 2023) introduces channel-level augmentations tailored to biosignals. Nevertheless, contrastive objectives can suffer from false negatives, especially in sleep staging, where adjacent segments may share labels. SSLAPP (Lee, Seong, and Chae 2022) addresses this by adversarially generating high-quality positives and performing attention-guided augmentation in latent space, mitigating semantic collisions in pair construction.

As view construction strategies matured, research attention gradually shifted toward the design of the encoder $\mathcal{F}(\cdot)$ to better capture EEG dynamics. Early approaches predominantly relied on CNN-based architectures (Mohsenvand, Izadi, and Maes 2020; Li and Metsis 2022; Kumar et al. 2022), which primarily model local temporal patterns. Recognizing the intrinsically non-stationary and multi-scale nature of neural signals, later works incorporate temporal-spectral duality into representation learning. TF-C (Zhang et al. 2022) enforces consistency between time-domain and frequency-domain embeddings of the same signal, explicitly coupling complementary signal views. More recently, transformer-based encoders have been introduced to support cross-dataset and cross-subject learning at scale. BIOT (Yang, Westover, and Sun 2023) employs a biosignal transformer to unify heterogeneous modalities, while LEAD (Wei et al. 2025) further regularizes representations through subject-level consistency, encouraging invariant embeddings across samples from the same individual.

Contrastive Predictive Coding Contrastive-based SSL defines learning objectives by comparing representations of paired inputs and contrasting them against a set of negatives. This paradigm can be expressed as:

$$\mathcal{L}_C = \mathcal{L}(\mathcal{G}(z_1), z_2, \mathcal{N}) \quad (9)$$

where $z_1 = \mathcal{F}_1(\mathcal{T}_{d,s}(\mathcal{X}_{src}, \mathcal{M}))$ and $z_2 = \mathcal{F}_2(\mathcal{T}_{d,s}(\mathcal{X}_{tgt}))$, $\mathcal{F}_1(\cdot)$ produces a query representation from the causal context while $\mathcal{F}_2(\cdot)$ generates the target embedding from a future segment. The projection head $\mathcal{G}(\cdot)$ maps representations into a latent space. The loss contrasts the positive pair (z_1, z_2) with negatives $\mathcal{N}$, promoting alignment of matched pairs and separation from mismatches. Inspired by BERT and wav2vec 2.0 (Baevski et al. 2020), Bendr (Kostas, Aroca-Ouellette, and Rudzicz 2021) encodes EEG segments into a unified sequence of learned vector representations. Maeeg (Chien et al. 2022) and GEFM (Wang, Suzumura, and Kanezashi 2024) have similar architecture to Bendr with different PT mindset. Although Bendr can extract features well, it ignores spatial information. To address this issue, several models have been proposed. GEFM adds a graph structure to Bendr and proposes a sequence length adjustment mechanism before GNN to make EEG signal lengths consistent for fixed-length node features in GNN. Zhu et al. (Zhu et al. 2023) incorporated spatial information through channel-mixing augmentation, effectively enhancing dataset diversity and improving the performance of the contrastive EEG2Vec framework. Another line of work extends CPC to better exploit multichannel structure and continual data

streams. MBrain (Cai et al. 2023) augments CPC with spatial awareness by aggregating multichannel semantics to predict single-channel local representations, encouraging implicit spatio-temporal dependencies across channels. BrainUICL (Zhou et al. 2025b) further combines CPC with replay-based continual learning, and the contrastive model is incrementally updated via joint training with replayed samples.

Cross-modal Contrast Inspired by multimodal contrastive learning, BFM s extend cross-modal alignment to EEG, where “modality” arises from the inherent heterogeneity of brain recordings. At the intra-signal level, different views of the same EEG (e.g., raw waveforms vs. time-frequency representations) can serve as distinct modalities. MuLEEG (Kumar et al. 2022) follows this paradigm by contrasting multi-view raw signals with spectrograms. Beyond view-level heterogeneity, MCSP (Wei et al. 2025) mines complementary information across modalities and models latent interactions within each domain, particularly spatial structure via graphs. For truly multi-sensor settings, SleepFM (Thapa et al. 2024) jointly embeds three biosignal modalities using both pairwise contrast and a leave-one-out objective, aligning each modality to the average of the others. Brant-X (Zhang et al. 2024a) further targets EEG-EXG coupling by aligning representations at both patch and sequence levels, capturing correlations at fine- and coarse-grained semantic scales. Moving to heterogeneous cross-modal alignment beyond biosignals, (Ferrante et al. 2024) align neural recordings with visual stimuli, demonstrating that visual content can be decoded from neural data and that images can be mapped into neural representation spaces.

Generative-based Methods

Generative-based SSL formulates pretraining objectives by reconstructing or predicting structured targets from transformed inputs, avoiding explicit pair construction and negative sampling (Zhang et al. 2023b). This paradigm has become increasingly prevalent in recent BFM s. Given a transformed input $\mathcal{T}_d(x)$, an encoder $\mathcal{F}_{\theta_1}(\cdot)$ produces a continuous latent representation $z_i = \mathcal{F}_{\theta_1}(\mathcal{T}_d(x_i))$ where $x_i \in \mathcal{X}_{src}$. A learnable codebook $\mathcal{Q}_{\theta_3}(\cdot)$ discretizes z_i into a latent embedding, which is then processed by a decoder $\mathcal{G}_{\theta_2}(\cdot)$ to reconstruct a target view or predict future content. The objective is defined as:

$$\mathcal{L}_G = \mathcal{L}(\mathcal{G}_{\theta_2}(\mathcal{Q}_{\theta_3}(\mathcal{F}_{\theta_1}(\mathcal{T}_d(x_i), \mathcal{M}))), \mathcal{F}_{\theta_1}(\mathcal{T}_d(x_i))) \quad (10)$$

By minimizing the reconstruction or prediction loss, the model learns latent representations that retain information necessary to recover the underlying signal structure.

Autoregressive The success of autoregressive (AR) language models in modeling long-range dependencies via next-token prediction has motivated the adoption in BFM s. Using a decoder-only transformer architecture, GPT-style models (Radford et al. 2019) have been adapted to EEG pretraining in several BFM s (Yue et al. 2024; Wang et al. 2024a; Lloyd et al. 2024). Early efforts such as Neuro-GPT (Cui et al. 2024) segment continuous EEG into fixed-length chunks and treat each chunk as a token, enabling a GPT model $\mathcal{G}_{\theta_2}(\cdot)$

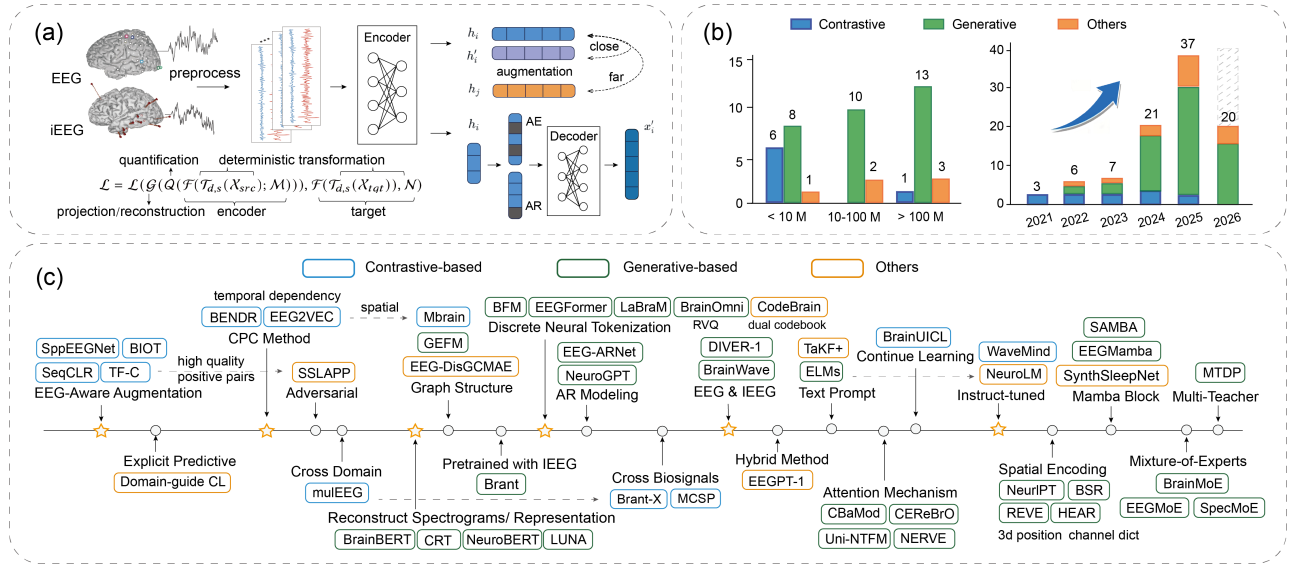


Figure 6: **Overview of BFMs.** (a) A unified pretraining pipeline. EEG/iEEG recordings are preprocessed, encoded into latent representations, and optimized under different SSL paradigms. (b) Model scale statistics. Parameter-size buckets by training paradigm are shown for the reported subset, using each model's maximum parameter count, alongside yearly model counts under the same grouping. (c) Timeline-style family tree of representative BFMs from 2021 to 2026.6, organized by paradigm and annotated with major methodological shifts.

to learn spatio-temporal structure by predicting masked segments. Building on this framework, Stress-GPT (Lloyd et al. 2024) fine-tunes the pretrained model for stress-related tasks, demonstrating the transferability of AR-pretrained representations.

Subsequent works refine the tokenization and prediction strategy to better reflect EEG-specific structure. EEGPT-2 (Yue et al. 2024) adopts an electrode-wise autoregressive formulation, treating each electrode signal x_i^e as a basic token and modeling temporal dependencies through an Electrode Temporal Encoder. EEG2Rep (Mohammadi Foumani et al. 2024) shifts target prediction into latent space via context-driven masking and introduces a semantic subsequence preserving mechanism to provide more informative masked inputs. Inspired by large language models, NeuroLM (Jiang et al. 2024) generalizes autoregressive pretraining to multichannel EEG, explicitly modeling inter-channel dependencies. More recently, LBLM (Zhou et al. 2025a) unifies temporal and spectral autoregression to capture spectro-temporal dynamics, while ECHO (Liu et al. 2025) constructs discrete, prompt-like context supports encoding hierarchical signal-task-label relations, improving in-context learning for downstream EEG tasks.

Autoencoder Autoencoder is the most common method in BFMs pretraining. To learn informative representations $\mathcal{M}(x)$, AE-based BFMs rely on carefully designed disturbance mechanisms that force reconstruction from partial observations. Masked Autoencoders (MAE) (He et al. 2022) randomly mask input regions and reconstruct missing content from visible context, encouraging the model to capture global structure. Inspired by both Bendr (Kostas, Aroca-

Ouellette, and Rudzicz 2021) and MAE, MAEEG (Chien et al. 2022) and GEFM (Wang, Suzumura, and Kanezashi 2024) adopt masked reconstruction for EEG pretraining. UniEEG (Jin et al. 2025) further introduces Masked Signal Modeling (MSM) with electrode-wise masking, while EEGPT-1 (Wang et al. 2024a) embeds local spatio-temporal patches as tokens and proposes a dual SSL objective combining mask-based reconstruction with spatio-temporal alignment.

Beyond masking strategies, modern AE-based BFMs primarily differ in the design of the encoder $\mathcal{F}_{\theta_1}(\cdot)$, which must model long-range temporal dependencies while capturing cross-channel interactions. Early models such as BrainBERT (Wang et al. 2023) largely encode electrodes independently, limiting spatial coupling. Subsequent works introduce stronger spatial inductive biases: EEG2TEXT (Liu et al. 2024) employs multi-view attention to reflect region-wise processing, while CBaMod (Wang et al. 2024b) adopts a criss-cross encoder that alternates spatial-temporal attention with asymmetric conditional positional encoding. To handle heterogeneous montages, LUNA (Döner et al. 2025) unifies variable electrode layouts into a fixed latent space via learned queries, and REVE (Ouahidi et al. 2025) injects anatomical priors through 4D positional embeddings derived from electrode coordinates and time indices. Beyond attention-based designs, Mamba-style encoders enable linear-time sequence modeling for long recordings. SynthSleepNet (Lee et al. 2025) introduces a Mamba-based Temporal Context Module for inter-epoch dependencies, while SAMBA (Hong, Mackellar, and Ghane 2025) proposes a Multi-head Differential Mamba to suppress background noise while aggregating contextual information.

However, the inherent randomness, non-stationarity, and nonlinearity of neurophysiological signals make direct amplitude reconstruction in the spatiotemporal domain suboptimal (Wu et al. 2022). As a result, recent BFM s increasingly perform masking and reconstruction in transformed spaces $\mathcal{T}_d(\cdot)$ or low-dimensional latent variables z_i . Neuro2vec (Wu et al. 2022) pioneers denoising in the Fourier domain, while CRT (Zhang et al. 2023b) performs cross-domain dropping–reconstruction to align time- and frequency-domain representations. Building on Fourier-domain MAE, NeuroBERT (Wu et al. 2024) introduces Fourier Inversion Prediction (FIP) as a pretraining objective. In parallel, spectrogram-based reconstruction has gained traction due to its rich time–frequency semantics, supporting both signal understanding and downstream learning (Wang et al. 2023; Ogg and Coon 2024; Yuan et al. 2024).

Most AE-based BFM s are developed for non-invasive EEG, while intracranial EEG (iEEG) remains comparatively underexplored. To bridge this gap, BrainBERT (Wang et al. 2023), Brant (Zhang et al. 2023a), and BrainWave (Yuan et al. 2024) extend AE-based pretraining to SEEG recordings. BrainBERT leverages SEEG data from subjects watching videos, Brant adopts a dual-encoder MAE to jointly capture temporal dependencies and spatial correlations, and BrainWave scales pretraining to over 16,000 subjects, setting new benchmarks for diagnosis tasks. While these approaches substantially improve performance, they often increase model size. To address this, CEReBrO (Dimofte et al. 2025) proposes a compact encoder that represents brain oscillations through alternating attention, offering a parameter-efficient alternative for AE-based BFM s.

Codebook The codebook mechanism, originally introduced by Vector-Quantized Variational Autoencoders (VQ-VAE) (Van Den Oord, Vinyals et al. 2017), offers a key advantage over standard VAEs by discretizing continuous representations into a finite set of tokens. Recently, this paradigm has been adapted to EEG and iEEG modeling, where encoder outputs are quantized into discrete codebook entries via a quantization operator $\mathcal{Q}_{\theta_3}(\cdot)$ prior to decoding. Such discretization enables token-based representations of neural signals, bridging continuous biosignals with sequence modeling frameworks. The codebook is optimized using the vector-quantization objective: $\mathcal{L}_{vq} = \|sg[\mathcal{Q}_{\theta_3}(z_i)] - z_i\|_2^2 + \beta \|sg[z_i] - \mathcal{Q}_{\theta_3}(z_i)\|_2^2$ where $sg[\cdot]$ denotes the stop-gradient operator. Compared to conventional masked reconstruction, vector-quantization provides a discrete interface for neural sequence modeling, although its downstream benefits depend on the tokenizer design and adaptation setting.

A central challenge of this paradigm lies in learning an effective quantization operator $\mathcal{Q}_{\theta_3}(\cdot)$. EEGFormer (Chen et al. 2024) first applies codebook-based discretization to EEG pretraining, generating discrete indices that yield transferable and interpretable representations. LaBraM (Jiang, Zhao, and Lu 2024) further introduces a neural codebook that quantizes patch-level embeddings, with the codebook optimized through reconstruction in the Fourier domain. BioSerenity-E1 (Bettinardi, Rahmouni, and Gimenez

2025) adopts a transformer-based VQ-VAE to tokenize EEG spectra and performs masked token prediction, forcing the model to capture complex spatiotemporal dependencies and achieving state-of-the-art diagnostic performance. Beyond reconstruction-based objectives, NeuroLM (Jiang et al. 2024) extends this line of work by introducing vector-quantized time–frequency prediction and aligning EEG tokens with textual representations via adversarial training. After discretization, it employs multichannel autoregressive modeling, enabling LLM s to predict EEG tokens in a manner analogous to language modeling.

While codebook learning is often instantiated within autoencoder-style frameworks, it should be viewed as a general representation mechanism rather than a method specific to AE-based BFM s. Discrete neural tokens can serve as a unifying interface across reconstruction, contrastive, and autoregressive paradigms, motivating their independent treatment in this appendix.

Other Advanced Methods

Beyond the SSL paradigms discussed above, BFM s explore explicit predictive objectives, hybrid formulations, and post-SSL instruction tuning. These methods extend the pretraining framework along dimensions of supervision, objective design, and task alignment.

Explicit predictive In the context of EEG, several predictive tasks have been specifically proposed by updating the form of $\mathcal{G}_{\theta_2}(\cdot)$ to better align with neurophysiological characteristics. Domain-guided CL (Wagh et al. 2021) introduces Behavioral State Estimation, which predicts the delta-beta ratio to capture arousal-related spectral patterns. The Stopped Band Prediction pretext task (Jo et al. 2023) focuses on learning frequency-aware representations by asking the model to identify which frequency band has been suppressed in the input signal to capture frequency information. And the Temporal Trend Identification task (Jo et al. 2023) aims to extract temporal dynamics by classifying pre-defined trends. Although these tasks are neurophysiologically interpretable, their dependence on specific assumptions about signal behavior has still limited their adoption in BFM s.

Hybrid-based Hybrid SSL combines multiple self-supervised objectives within a unified framework to exploit complementary learning signals (Weng et al. 2025). In EEG, Domain-guided contrastive learning (Wagh et al. 2021) designs SSL pretext tasks grounded in neurophysiological priors, leveraging similarity of brain activity as well as behavioral-state and age-related differences. EEG-DisGCM AE (Wei et al. 2024) further introduces a graph-structured hybrid that couples Graph Contrastive pretraining with Graph MAE. LCM (Chen, Chen, and Tsai 2025) follows a different route, using masked generative reconstruction as the primary objective and adding a momentum-updated target encoder with a contrastive term to stabilize representation learning. More broadly, BFM s increasingly pair contrastive objectives with generative modeling to jointly enforce cross-view consistency and preserve signal structure (Wang et al. 2024a; Chen, Chen, and Tsai 2025; Li et al. 2025a; Lee

et al. 2025). Other hybrids integrate reconstruction with autoregressive prediction to capture both local structure and long-range temporal dynamics (Jiang et al. 2025; Yang et al. 2025), and may further incorporate adversarial objectives for domain-aware alignment (Jiang et al. 2024; Lee, Seong, and Chae 2022).

Instruction-tuned Recent instruction-tuned BFM bridge EEG representations with language semantics through lightweight alignment modules while keeping LLM backbones mostly frozen. NeuroLM (Jiang et al. 2024) discretizes EEG into temporal–frequency tokens via VQ encoding and aligns EEG and text embeddings through domain-adversarial learning, followed by multichannel autoregressive pretraining and prompt-based multi-task instruction tuning. TaKF+ (Jeon et al. 2025) emphasizes parameter-efficient transfer by injecting task information through adapters and cross-attention blocks after masked patch pretraining. UniMind (Lu et al. 2025) proposes a Neuro-Language Connector that extracts task-relevant temporal–spatial semantics using learnable queries with task-aware routing, and aligns them to the LLM latent space for instruction-conditioned generation. WaveMind (Zeng et al. 2025) combines contrastive multimodal alignment with CLIP and instruction tuning on a large EEG instruction dataset, enabling conversational EEG understanding across tasks.